



Kobayashi-hitchin correspondence for saturated reflexive parabolic sheaves on Kähler manifolds

Tianshu Jiang¹ · Jiayu Li^{1,2}

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Abstract

In this paper, we study the Kobayashi-Hitchin correspondence in the setting of parabolic sheaves with a simple normal crossing divisor over a compact Kähler manifold using the method of Hermitian-Yang-Mills flow.

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✉ Tianshu Jiang
jts2021@mail.ustc.edu.cn
Jiayu Li
jiayuli@ustc.edu.cn

¹ School of Mathematical Sciences, University of Science and Technology of China, Hefei 230026, P.R. China

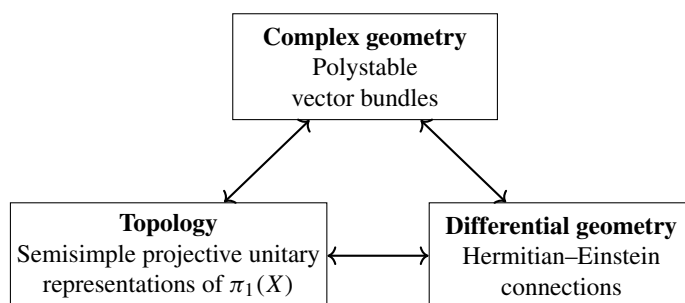
² AMSS, CAS, Beijing 100080, P.R. China

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1 Introduction

1.1 Background

The concept of Kobayashi-Hitchin correspondence is developed from an insightful discovery of M. S. Narasimhan and C. S. Seshadri [30] which reveals the correspondence between the irreducible unitary representations of the fundamental group of a compact Riemann surface and stable holomorphic vector bundles living on it with vanishing degree. With successive contributions from S. K. Donaldson [10, 11], S. Kobayashi [20, 21], Lübke [25], Hitchin [16], Uhlenbeck and Yau [37], etc., it takes into its modern shape which achieves the Trinity on a compact Kähler manifold X as is illustrated by the following diagram.



When we study the moduli space of the stable vector bundles over X , the correspondence enables the moduli space to interweave rich geometric structures from different perspectives.

Another direction of generalization is to consider the correspondence problem in the quasiprojective setting. It was pioneered by V. Mehta and C. S. Seshadri [27] when they studied the unitary representations of the fundamental group of a punctured Riemann surface X° , as a generalization of Narasimhan and Seshadri's [30] work. In their paper, they established the correspondence between the unitary representation of the fundamental group of X° and the concept of stable parabolic bundle over its completion X . Inspired by their work, the concept of parabolic vector bundles was gradually generalized to higher dimensional cases. For the purpose of illustration, let us consider the simplest case.

Let (X, ω) be a compact Kähler manifold and D be a smooth irreducible divisor. A parabolic bundle is a tuple $F_* = (F, \mathbf{F})$ consisting of an underlying holomorphic vector bundle $F \rightarrow X$ and a parabolic structure $\mathbf{F}$ along the divisor. The parabolic structure $\mathbf{F}$ is a descending filtration of holomorphic subbundles of $F|_D$:

$$F|_D = F_0 \supsetneq F_1 \supsetneq F_2 \supsetneq \cdots \supsetneq F_\ell \supsetneq 0$$

with weights $0 \leq a_i < a_{i+1} < 1$ assigned to F_i . We use the subscript “*” to emphasize the parabolic structure. And if we omit the “*” and just write F , we mean the underlying bundle of the parabolic bundle. As in the case of an ordinary vector bundle, associated to a parabolic bundle, we also have the parabolic versions of the Chern characters, the degrees, the slope stability with respect to ω , etc. There are other characterizations of parabolic bundles provided, e.g., in [5, 17, 28]. More generally, the notions of parabolic sheaves with a general divisor are also available. It was firstly introduced by M. Maruyama and K. Yokogawa [26]

to study the moduli problem. But in our paper, we follow the definitions of T. Mochizuki provided in [29, Chapter 3]. See also Section 2 for the details.

Parabolic bundles have attracted considerable attention since the 1990s. It is much more important and fascinating to study a parabolic bundle equipped with a Higgs field which has logarithmic singularities along the divisor. As it will lead to the nonabelian Hodge correspondence for quasiprojective smooth varieties pioneered by C. T. Simpson [33]. This subject tries to replace the objects in the above Trinity with semisimple local systems, harmonic bundles and stable parabolic Higgs bundles with trivial Chern classes over a quasiprojective variety. J. Jost and K. Zuo [18] built the bridge for the first two objects and T. Mochizuki [29] built the bridge for the last two. O. Biquard [3] dealt with the case when the base space is a Kähler manifold with a smooth divisor. The problem is still open if the setting is a Kähler manifold with a simple normal crossing divisor. Nowadays, this subject is still active with researchers trying to generalize the settings to sheaves over spaces with certain singularities, e.g., Klt, Dlt varieties. When there is no Higgs field, this problem was also considered by, e.g., [22, 23, 35] in different settings.

In particular, the second author [22] proved that in the setting of a compact Kähler manifold (X, ω) with a simple normal crossing divisor D , a parabolic bundle F_* is stable with respect to ω if and only if there exists a Hermitian-Einstein metric H_δ on $F|_{X \setminus D}$ with respect to a conical Kähler metric ω_δ (a metric with mild singularities near D), moreover H_δ is compatible with the parabolic structure. As a consequence, the author obtained the parabolic version of the Bogomolov-Gieseker inequality.

1.2 Purpose

It is worth noting that although the conical metric ω_δ used by the second author are in the same cohomology class of ω in the sense of current, it is not very satisfactory to only get a Hermitian-Einstein metric with respect to ω_δ rather than ω . The purpose of our paper is to generalize the previous results to the setting of semistable parabolic sheaves and in the mean time fix the above issue. We give a brief description of our main results in the following.

Let $\mathcal{F}_*$ be a saturated reflexive parabolic sheaf of rank r associated to (X, ω, D) . We put $\text{ch}_k(\cdot)$ as the k -th Chern character operator of a parabolic sheaf. As usual, we define the degree $\deg_\omega(\mathcal{F}_*)$ of a parabolic sheaf $\mathcal{F}_*$ as

$$\deg_\omega(\mathcal{F}_*) := \text{ch}_1(\mathcal{F}_*) \cdot \frac{[\omega]}{(n-1)!}$$

and the slope $\mu_\omega(\mathcal{F}_*)$ as

$$\mu_\omega(\mathcal{F}_*) := \frac{\deg_\omega(\mathcal{F}_*)}{\text{rank}(\mathcal{F}_*)}.$$

We say that $\mathcal{F}_*$ is μ_ω -stable (or simply ω -stable) (resp. semistable) with respect to the Kähler metric ω , if for any proper parabolic subsheaf $\mathcal{S}_*$ we have

$$\mu_\omega(\mathcal{S}_*) < (\text{resp. } \leq) \mu_\omega(\mathcal{F}_*).$$

From Section 2, we know that $\mathcal{F}_*$ is a locally abelian parabolic bundle outside an analytic subset W of codimension at least 3. We put $X^\circ := X - W$ and $F_* := \mathcal{F}_*|_{X^\circ}$. In the rest of the paper, we may sometimes simply call F the regular part of $\mathcal{F}$ and $F|_{X^\circ \setminus D}$ the regular part of $\mathcal{F}_*$.

Definition 1.1 We say that an Hermitian metric H on $F|_{X^\circ \setminus D}$ is compatible with the parabolic sheaf $\mathcal{F}_*$ (or with its parabolic structure) if the following conditions are satisfied

1. $\frac{\sqrt{-1}}{2\pi} \text{Tr}(F_H)$ represents $\text{ch}_1(\mathcal{F}_*)$ where F_H is the Chern curvature tensor.
2. For any proper parabolic subsheaves $\mathcal{S}_*$ of $\mathcal{F}_*$, let $H|_{\mathcal{S}_*}$ be the induced metric on the regular part of $\mathcal{S}_*$, we have $\text{ch}_1(\mathcal{S}_*) \leq \frac{\sqrt{-1}}{2\pi} \text{Tr}(F_{H|_{\mathcal{S}_*}})$ in the sense of current.

And it is admissible if

- $F_H \in L^2(H, \omega)$.
- $\Lambda F_H \in L^\infty(H)$ where ΛF_H is the contraction of F_H with respect to ω .

Remark 1.1 The inequality in condition 2 could be strengthened to equality when $\mathcal{F}_*$ is a parabolic bundle. But in the case of sheaves, this is the best we can expect.

The concept of admissible metrics on reflexive sheaves was first introduced by S. Bando and Y. Siu in [2] where the Kobayashi-Hitchin problem of reflexive sheaves was concerned.

Definition 1.2 A Hermitian metric H on a vector bundle F over a Kähler manifold (X, ω) is a Hermitian-Einstein (H-E) metric if $\sqrt{-1}\Lambda_\omega F_H - \lambda \cdot \text{id}_F = 0$ all over X where λ is a real constant and Λ_ω is the adjoint of the Lefschetz operator.

Definition 1.3 A family of Hermitian metrics $H(t)$ parameterized by $t \in [0, \infty[$ on a vector bundle F over a Kähler manifold (X, ω) is approximate Hermitian-Einstein (H-E) if

$$\lim_{t \rightarrow \infty} \|\sqrt{-1}\Lambda_\omega F_{H(t)} - \lambda \cdot \text{id}_F\|_{L^\infty(X, H(t))} = 0.$$

We obtain the following theorems:

Theorem 1.2 (Theorem 6.14) *A saturated reflexive parabolic sheaf $\mathcal{F}_*$ over (X, ω, D) is μ_ω -polystable if and only if there exists an admissible H-E metric with respect to ω on $F|_{X^\circ \setminus D}$ which is compatible with $\mathcal{F}_*$.*

Theorem 1.3 (Theorem 6.15) *A saturated reflexive parabolic sheaf $\mathcal{F}_*$ over (X, ω, D) is μ_ω -semistable if and only if there exists a family of admissible approximate H-E metrics with respect to ω on $F|_{X^\circ \setminus D}$ which are compatible with $\mathcal{F}_*$.*

As a consequence, we obtain the parabolic version of the Bogomolov-Gieseker inequality. Let us denote

$$\Delta(\mathcal{F}_*) := \frac{\text{ch}_1(\mathcal{F}_*)^2}{2 \text{rank}(\mathcal{F}_*)} - \text{ch}_2(\mathcal{F}_*)$$

as the Bogomolov-Gieseker discriminant.

Corollary 1.4 (Corollary 6.16) *If $\mathcal{F}_*$ is μ_ω -semistable with respect to ω , then*

$$\Delta(\mathcal{F}_*) \cdot [\omega]^{n-2} \geq 0.$$

Moreover, if $\mathcal{F}_$ is polystable, then the equality holds if and only if $\mathcal{F}|_{X \setminus D}$ is a vector bundle which admits a projectively flat H-E connection compatible with the parabolic structure.*

Establishing the Bogomolov-Gieseker inequality for sheaves gives us more flexibility in applications. As a consequence, we also prove the following result.

Theorem 1.5 (Theorem 7.1) *Let $\mathcal{F}_*$ be a saturated reflexive parabolic sheaf over a compact Kähler manifold (X, ω) which is semistable with respect to a nef and big class $[\eta]$. Then the Bogomolov-Gieseker inequality with respect to $[\eta]$ holds, i.e.,*

$$\Delta(\mathcal{F}_*) \cdot [\eta]^{n-2} \geq 0.$$

1.3 Outline of the paper

- §2: We introduce the abelian category whose objects are parabolic sheaves. Roughly speaking, a parabolic sheaf $\mathcal{F}_*$ is a decreasing filtration of sheaves which only degenerate on a divisor D . We discuss the notions of the morphisms, the Chern characters, the degrees, the stability condition, etc. We show that a saturated reflexive parabolic sheaf $\mathcal{F}_*$ is a parabolic bundle outside an analytic set W with codimension at least 3.
- §3: We resolve the singularities of $\mathcal{F}_*$ by blowing up successively along the smooth centers above W . In summary, we will have a modification $\pi : \tilde{X} \rightarrow X$. Let $\mathcal{E}$ be the exceptional divisor. To simplify notation, we use the same notation for objects pulled back by π . Then, over $\tilde{X}$, $\mathcal{F}$ can be embedded into a locally free sheaf E which is isomorphic to $\mathcal{F}$ outside $\mathcal{E}$. Then by blowing up further, E will inherit the parabolic structure from $\mathcal{F}_*$. We denote the parabolic bundle by E_* .
- §4: In order to use the Hermitian-Yang-Mills flow to deform the metric on the regular part of $\mathcal{F}_*$, we first constructed a family of conical Kähler metrics $\omega_{\epsilon\delta}$ on $\tilde{X} - D$ which is derived from $\pi^*\omega$. Particularly, such a family of metrics satisfies a Sobolev type inequality with a uniform constant. Accordingly, we construct a singular Hermitian metric $\hat{H}$ on E_* which vanishes on the proper transformation of D . Then we show that the curvature tensor of $\hat{H}$ will give us the first and second Chern characters of E_* as in the case of ordinary vector bundles. We believe that the parabolic metric we have constructed will give us all of the parabolic Chern characters. It also induces a metric on the regular part of $\mathcal{F}_*$ since $\tilde{X}$ and X are isomorphic outside W . As one can expect, the curvature tensor of $\hat{H}$ will give us the first and second Chern characters of $\mathcal{F}_*$.
- §5: We recall the concept of the analytic stability of a vector bundle with respect to a Hermitian metric on the bundle and a Kähler form on the base manifold. More importantly, we show that the analytic stability of the Hermitian holomorphic bundle $(F|_{X \setminus D}, \hat{H})$ is equivalent to the stability of the parabolic sheaf $\mathcal{F}_*$.
- §6: We analyse a family of Hermitian-Yang-Mills (H-Y-M) flows with respect to the conical Kähler metrics $\omega_{\epsilon\delta}$ and the initial metric $\hat{H}$ on the bundle $E|_{\tilde{X}-D}$. More precisely, we consider the following family of H-Y-M flows:

$$\begin{cases} H^{-1} \cdot \frac{dH}{dt} = -2 \left(\sqrt{-1} \Lambda_{\omega_{\epsilon\delta}} F_H - \lambda_{\epsilon\delta} \cdot \text{id}_F \right) \\ \det(H) = \det(\hat{H}) \\ H(0) = \hat{H} \end{cases}$$

We show that as ϵ and δ tends to 0, the family of flows will converge to the flow

$$\begin{cases} H^{-1} \cdot \frac{dH}{dt} = -2 \left(\sqrt{-1} \Lambda_{\omega} F_H - \lambda \cdot \text{id}_F \right) \\ \det(H) = \det(\hat{H}) \\ H(0) = \hat{H} \end{cases}$$

living on the regular part of $\mathcal{F}_*$ where

$$\lambda := \frac{2\pi \cdot \mu_{\omega}(\mathcal{F}_*)}{\text{Vol}(X, \omega)}.$$

Then we will show that under the stability (resp. semistability) condition, the flow will converge to an admissible H-E metric (resp. a family of admissible approximate H-E

metrics) on $F|_{X^\circ \setminus D}$ which is compatible with $\mathcal{F}_*$. We will also prove the converse part where the admissibility of the H-E metric is crucial.

- §7: As an application, we prove the Bogomolov-Gieseker type inequality for a semistable parabolic sheaf with respect to a big and nef class using Jordan-Hölder filtration.

2 Elementary notions on parabolic sheaves

We review the elementary notions on parabolic sheaves. These concepts can be found, e.g., in [29, Chapter 3], [17, Section 2], [26, Section 1]. The definitions of parabolic sheaves introduced in these papers differ slightly but are all similar in spirit. Indeed, it was discussed in [17] that the diversified definitions appeared in the literature are equivalent when restricted to locally abelian parabolic bundles. In this paper, we basically follow the definitions introduced in [29].

Let (X, D) be a pair of complex manifold X and $D = \bigcup_{i \in I} D_i$ a simple normal crossing divisor. We impose a linear order on I . We introduce parabolic bundles associated to (X, D) first and then do the generalizations to sheaves. If we say a generic property P holds in codimension $k - 1$, we mean that P holds outside an analytic subset of codimension k .

2.1 Parabolic bundles

We start with the notion of compatible filtrations. Fix a vector space V .

Definition 2.1 A decreasing left continuous filtration of V is a set of subspaces $F = \{F_a | a \in [0, 1[\}$ indexed by $a \in [0, 1[$ such that:

1. $F_a \supseteq F_b$ for $a \leq b$,
2. $F_0 = V$ and $F_b = 0$ for b sufficiently close to 1,
3. If ϵ is sufficiently small, then $F_{a-\epsilon} = F_a$.

All filtrations appear in this paper will be decreasing and left continuous hence we will simply call them filtrations in the rest of the paper.

Given a tuple of filtrations $F = \{^i F | i \in I\}$. For any $J \subset I$, $\eta \in [0, 1[^J$, we put as follows:

$$^J F_\eta = \bigcap_{j \in J} ^j F_{\eta_j}$$

Definition 2.2 A tuple of filtrations $F = \{^i F | i \in I\}$ is compatible if they admit a common splitting, i.e. a decomposition $V = \bigoplus_{\eta \in [0, 1[^I} U_\eta$ such that:

$$^I F_\rho = \bigoplus_{\eta \geq \rho} U_\eta$$

For a coherent sheaf $\mathcal{V}$ over a complex manifold M , we define a decreasing left continuous filtration of subsheaves in the same manner. A filtration of subsheaves $\mathcal{F} = \{\mathcal{F}_a | a \in [0, 1[\}$ is called in the category of vector bundles, if all the subsheaves are locally free and $\text{Gr}_a := \mathcal{F}_a / \mathcal{F}_{>a}$ is locally free for all a .

Definition 2.3 A tuple of vector bundle filtrations $F = \{^i F | i \in I\}$ is compatible if:

1. For any $x \in X$, F_x is compatible in the sense of Definition 2.2,

2. For any $J \subset I$ and $\rho \in [0, 1[$, ${}^J F_\rho$ is a subbundle.

Let F be a vector bundle over X . On each irreducible component D_i , one can specify a vector bundle filtration ${}^i F$ of $F|_{D_i}$. Hence one obtain a tuple of filtrations $\mathbf{F} = \{{}^i F | i \in I\}$. For any $J \subset I$, we put $D_J = \bigcap_{j \in J} D_j$. Then ${}^J \mathbf{F}$ is a tuple of filtrations of the vector bundle $F|_{D_J}$.

Definition 2.4 The tuple $(F, \mathbf{F})$ given as above is called a parabolic bundle, if for all $J \subset I$, restricting to any irreducible component of D_J , ${}^J \mathbf{F}$ is compatible in the sense of Definition 2.3. $\mathbf{F}$ is called the parabolic structure on F .

Definition 2.5 A parabolic bundle $(F, \mathbf{F})$ is called locally abelian, if for each $x \in X$, there is a neighborhood U_x of x such that $(F, \mathbf{F})|_{U_x}$ is isomorphic to a direct sum of parabolic line bundles in the category of parabolic sheaves (see the definition in the next subsection).

N. Borne [6] shows

Theorem 2.1 Given a parabolic bundle $(F, \mathbf{F})$ over (X, D) . If all of the intersections of the subsheaves ${}^i \mathcal{F}_a$ defined by the exact sequence

$$0 \rightarrow {}^i \mathcal{F}_a \rightarrow F \rightarrow F|_{D_i}/{}^i F_a \rightarrow 0$$

are locally free, then $(F, \mathbf{F})$ is locally abelian.

Remark 2.2 If the base space is a smooth algebraic variety, Borne's theorem is stronger. It says that the parabolic bundle is isomorphic to a direct sum of parabolic line bundles in any Zariski neighborhood.

2.2 Parabolic sheaves

Let $\mathcal{F}$ be a torsion-free coherent sheaf of $\mathcal{O}_X$ -modules.

Definition 2.6 A parabolic structure on $\mathcal{F}$ is a tuple $\mathcal{F} = \{{}^i \mathcal{F} \mid i \in I\}$ of decreasing left continuous filtrations ${}^i \mathcal{F}_a$ indexed by $a \in [0, 1[$ with finite length l_i such that ${}^i \mathcal{F}_a \supset \mathcal{F}(-D_i)$ for any $a \in [0, 1[$ and ${}^i \mathcal{F}_a = \mathcal{F}(-D_i)$ if a is sufficiently large.

For the filtration ${}^i \mathcal{F}$ over D_i , let ${}^i a = \{{}^i a_0, {}^i a_1, \dots, {}^i a_{l_i}\}$ be the increasing sequence of indexes such that ${}^i \text{Gr}_{i_{a_k}} \mathcal{F}_* \neq 0$. This is often called the weights of the parabolic structure on D_i . We call $\mathcal{F}$ the top flag and $\mathcal{F}(-D_i)$ the handle in the filtration ${}^i \mathcal{F}$.

Definition 2.7 A parabolic sheaf is a tuple $\mathcal{F}_* = (\mathcal{F}, \mathcal{F})$ consisting of a underlying sheaf $\mathcal{F}$ and a parabolic structure $\mathcal{F}$ on $\mathcal{F}$.

Definition 2.8 A parabolic subsheaf of $\mathcal{F}_* = (\mathcal{F}, \mathcal{F})$ is defined as a quotient torsion-free subsheaf $\mathcal{E}$ with the naturally induced parabolic structure.

Definition 2.9 A morphism $f : \mathcal{F}_* \rightarrow \mathcal{G}_*$ is defined as a morphism of sheaves $f : \mathcal{F} \rightarrow \mathcal{G}$ such that $f({}^i \mathcal{F}_a) \subseteq {}^i \mathcal{G}_a$ for all $a \in [0, 1[$.

We denote the set of the morphisms between two parabolic sheaves $\mathcal{F}_*$ and $\mathcal{G}_*$ as $\text{Hom}(\mathcal{F}_*, \mathcal{G}_*)$.

Definition 2.10 A complex

$$\dots \rightarrow \mathcal{E}_* \rightarrow \mathcal{F}_* \rightarrow \mathcal{G}_* \rightarrow \dots$$

is exact at $\mathcal{F}_*$, if and only if

$$\cdots \rightarrow {}^i\mathcal{E}_a \rightarrow {}^i\mathcal{F}_a \rightarrow {}^i\mathcal{G}_a \rightarrow \cdots$$

is exact at ${}^i\mathcal{F}_a$ for all i and a .

A parabolic bundle gives rise to a parabolic sheaf in the following way. Given a parabolic bundle F_* , one define the parabolic structure ${}^i\mathcal{F}_a$ by the exact sequence

$$0 \rightarrow {}^i\mathcal{F}_a \rightarrow F \rightarrow F|_{D_i}/{}^iF_a \rightarrow 0.$$

Conversely, how far away is a parabolic sheaf from a parabolic bundle? The following lemmas give a partial answer.

We say that a parabolic sheaf $\mathcal{F}_*$ is reflexive if the underlying sheaf $\mathcal{F}$ is reflexive. And $\mathcal{F}_*$ is called saturated if $\mathcal{F}/{}^i\mathcal{F}_a$ are torsion-free $\mathcal{O}_{D_i}$ -modules for any i and a . We put

$${}^i\mathcal{F}_{a|D_J} := \mathfrak{Z}({}^Jm^*({}^i\mathcal{F}_a) \rightarrow {}^Jm^*(\mathcal{F}))$$

where Jm is the embedding of D_J .

Lemma 2.3 *All the intersections of subsheaves belonging to the parabolic structure of a saturated reflexive parabolic sheaf are reflexive.*

Proof T. Moichizuki [29, Proposition 3.1.2] shows that all the subsheaves ${}^i\mathcal{F}_a$ are reflexive. Then the lemma follow from the fact that the intersection of two reflexive subsheaves of a reflexive sheaf is reflexive. $\square$

Lemma 2.4 *A saturated reflexive parabolic sheaf is a locally abelian parabolic bundle in codimension 2.*

Proof Denote the sheaf by $\mathcal{F}_*$. Since $\mathcal{F}$ is reflexive, $\mathcal{F}$ is locally free outside an analytic set Z_0 with codimension 3. On each D_i , since $\mathcal{F}/{}^i\mathcal{F}_a$ are torsion-free $\mathcal{O}_{D_i}$ -modules, they are locally free outside an analytic subset Z_i of codimension 2 in D_i . Hence outside $\bigcup_{\{0,1\}} Z_i$, ${}^i\mathcal{F}|_{D_i}$ is a vector bundle filtration for all i . It is easy to check that for $i \neq j$, the vector bundle filtrations ${}^i\mathcal{F}|_{D_{ij}}$ and ${}^j\mathcal{F}|_{D_{ij}}$ are compatible outside an analytic subset Z_{ij} of codimension 2 in D_{ij} . Hence if we further delete the union of Z_{ij} and D_J with $|J| \geq 3$, $\mathcal{F}_*$ will be a parabolic bundle. By the above lemma, all of the intersections of ${}^i\mathcal{F}_a$ are reflexive, hence they are locally free outside an analytic subset Z' of codimension 3. Deleting Z' makes $\mathcal{F}_*$ locally abelian. $\square$

Lemma 2.5 *A parabolic sheaf admits a unique reflexive saturation which is isomorphic to the former in codimension 1, i.e. given a parabolic sheaf $\mathcal{F}_*$, there exists a unique saturated reflexive parabolic sheaf $\mathcal{F}'_*$ and a monomorphism $m : \mathcal{F}_* \rightarrow \mathcal{F}'_*$ which is an isomorphism in codimension 1.*

Proof Let $\mathcal{F}'$ be the double dual of $\mathcal{F}$. They are isomorphic outside an analytic subset Z of codimension 2. Since $\mathcal{F}'$ is normal, we define a parabolic structure $\mathcal{F}'^1$ as the unique extension of $\mathcal{F}|_{X-Z}$ in $\mathcal{F}'$. The subsheaves in $\mathcal{F}'^1$ are coherent due to Theorem 2.2 of [34]. Moreover, the top flag of ${}^i\mathcal{F}$ is $\mathcal{F}'$ and the handle is $\mathcal{F}'(-D_i)$. Hence $\mathcal{F}'/{}^i\mathcal{F}_a^1$ can be regarded as a sheaf of $\mathcal{O}_{D_i}$ -modules. Finally we construct ${}^i\mathcal{F}'_a$ as the inverse image of the $\mathcal{O}_{D_i}$ -torsion subsheaf of $\mathcal{F}'/{}^i\mathcal{F}_a^1$ under the quotient map. One readily verifies that $\mathcal{F}'_* = (\mathcal{F}', \mathcal{F}'^1)$ serves as an candidate. The uniqueness follows immediately from Lemma 2.3 and the fact that a reflexive sheaf is normal. $\square$

2.3 Parabolic Chern character

From now on, we assume that X is a compact Kähler manifold.

Definition 2.11 The parabolic Chern character $\text{ch}(\mathcal{F}_*)$ of a parabolic sheaf $\mathcal{F}_*$ is given by the formula

$$\text{ch}(\mathcal{F}_*) = \frac{\text{ch}(D) \int_{a_1=0}^1 \cdots \int_{a_k=0}^1 e^{\sum_{i=1}^k a_i \cdot [D_i]} \cdot \text{ch}({}^I \mathcal{F}_a)}{\int_{a_1=0}^1 \cdots \int_{a_k=0}^1 e^{\sum_{i=1}^k a_i \cdot [D_i]}} \quad (1)$$

where k is the cardinality of I and $\mathbf{a} \in [0, 1]^k$.

Remark 2.6 In this paper, we use the Chern character functor for an ordinary coherent analytic sheaf over a compact complex manifold provided in [19] where they generalized the intersection theory (more precisely, the GRR theorem) in the smooth algebraic setting to the smooth complex analytic setting. The above definition of the parabolic Chern character is based on their definition for an ordinary sheaf. However it is by no means standard. The above formula was originally obtained by Ivy and Simpon in [17] for locally abelian parabolic bundles. They proved the equivalence between the category of locally abelian parabolic bundles with rational weights over a pair (X, D) of a smooth scheme and a simple normal crossing divisor and the category of vector bundles over an associated Deligne-Mumford stack Z . Then they used the equivalence to define the parabolic Chern character of a locally abelian parabolic bundle and calculated the explicit formula as above. But for more general parabolic sheaves or even parabolic bundles, there is no hint for the above formula to hold. We use it as a definition for the following reasons.

Firstly, such a definition of Chern character has good functorial properties. For instance, given a short exact sequence of parabolic sheaves

$$0 \rightarrow \mathcal{E}_* \rightarrow \mathcal{F}_* \rightarrow \mathcal{G}_* \rightarrow 0,$$

we have $\text{ch}(\mathcal{F}_*) = \text{ch}(\mathcal{E}_*) + \text{ch}(\mathcal{G}_*)$. Indeed this is all we need for the purpose of proving the Bogomolov-Gieseker inequality in Section 7.

More importantly, it was calculated by Taher in [36] that for a locally abelian parabolic bundle in codimension 1 (resp. 2), we have following more concrete formulas for the first (resp. second) Chern character.

Suppose $\mathcal{F}_*$ is a locally abelian parabolic bundle in codimension 2, e.g. a saturated reflexive parabolic sheaf. We put ${}^i \text{Gr} \mathcal{F}_* := \bigoplus_{a \in [0, 1]} ({}^i \text{Gr}_a \mathcal{F}_* := {}^i \mathcal{F}_a / {}^i \mathcal{F}_{>a})$ as a graded sheaf of $\mathcal{O}_{D_i}$ -modules. On any irreducible component P of $D_i \cap D_j$, we put

$${}^P \text{Gr}_{(a_i, a_j)} \mathcal{F}_* := {}^P F_{a_i, a_j} / \sum_{x > a_i, y > a_j} {}^P F_{x, y}.$$

Then the formula reads as follows:

$$\begin{aligned} \text{ch}_1(\mathcal{F}_*) &= \text{ch}_1(\mathcal{F}) + \sum_{i \in I} \sum_{a \in [0, 1]} a \cdot \text{rank}_{D_i}({}^i \text{Gr}_a \mathcal{F}_*) \cdot [D_i] \\ \text{ch}_2(\mathcal{F}_*) &= \text{ch}_2(\mathcal{F}) + \sum_{i \in I} \sum_{a \in [0, 1]} a \cdot m_{i*} \left(\text{ch}_1({}^i \text{Gr}_a \mathcal{F}_*) \right) \\ &\quad + \frac{1}{2} \sum_{i \in I} \sum_{a \in [0, 1]} a^2 \cdot \text{rank}_{D_i}({}^i \text{Gr}_a \mathcal{F}_*) \cdot [D_i]^2 \end{aligned} \quad (2)$$

$$+ \sum_{\substack{i < j \\ (i,j) \in I^2}} \sum_{\substack{P \in Irr(D_i \cap D_j) \\ (a_i, a_j) \in [0,1]^2}} a_i \cdot a_j \cdot \text{rank}_P^P \text{Gr}_{(a_i, a_j)} \cdot [P]. \quad (3)$$

Remark 2.7 The above formula of the first Chern character was firstly introduced by V. Metha and C.S.Seshadri [27] on algebraic curves and then generalized to parabolic sheaves in higher dimensions by M. Maruyama and K. Yokogawa [26]. The formula of the second Chern character for a locally abelian parabolic bundles was obtained by the second author [22, Lemma 7.3, 7.4] using the curvature tensor of some adapted metric which degenerates on the divisor. And then T.Mochizuki [29] used it as a definition for saturated reflexive parabolic sheaves. After Iyer and Simpson [17] gave a general definition of parabolic Chern character for locally abelian parabolic bundles, Taher showed in [36] that the new general definition agrees with the classical ones.

Since in this paper, we only study the reflexive saturated parabolic sheaves, the Chern character functor given in the very beginning of this subsection serves as a good candidate. Next, we compare the Bogomolov-Gieseker discriminant of a parabolic sheaf with that of its reflexive saturation in the following lemma.

Lemma 2.8 *Let $\mathcal{F}'_*$ be the reflexive saturation of a parabolic sheaf $\mathcal{F}_*$. Then $\Delta(\mathcal{F}'_*) \geq \Delta(\mathcal{F}_*)$.*

Proof On the one hand, since they are isomorphic in codimension 1, $\mathcal{T}_a = {}^I\mathcal{F}'_a / {}^I\mathcal{F}_a$ is a torsion sheaf supported on an analytic subset of codimension 2. By [19, Theorem 1.1], $\text{ch}_1(\mathcal{T}_a)$ vanishes and $\text{ch}_2(\mathcal{T}_a)$ is a current represented by a nonnegative linear sum of irreducible components of $\text{supp } \mathcal{T}_a$ with codimension 2. On the other hand, by equation (1), we have

$$\text{ch}(\mathcal{F}'_*) - \text{ch}(\mathcal{F}_*) = \frac{\text{ch}(D) \int_{a_1=0}^1 \cdots \int_{a_k=0}^1 e^{\sum_{i=1}^k a_i \cdot [D_i]} \cdot \text{ch}(\mathcal{T}_a)}{\int_{a_1=0}^1 \cdots \int_{a_k=0}^1 e^{\sum_{i=1}^k a_i \cdot [D_i]}}.$$

Then it is easy to observe that $\text{ch}_1(\mathcal{F}'_*) = \text{ch}_1(\mathcal{F}_*)$ and $\text{ch}_2(\mathcal{F}'_*) - \text{ch}_2(\mathcal{F}_*)$ is a current represented by a nonnegative linear sum of irreducible components of $\text{supp } \mathcal{T}_a$ with codimension 2. Hence $\Delta(\mathcal{F}'_*) \geq \Delta(\mathcal{F}_*)$. $\square$

The notions of stability, polystability and semistability for parabolic sheaves are defined in the same way as for an ordinary sheaf. The existence and uniqueness of Harder-Narasimhan filtration and Jordan-Hölder filtration can be proved in a tautological way as for an ordinary sheaf. Also as in the case of an ordinary sheaf,

Lemma 2.9 *If a parabolic sheaf $\mathcal{F}_*$ is stable with respect to a class $[\eta] \in H^2(X, \mathbb{R})$, then its reflexive saturation $\mathcal{F}'_*$ is also η -stable.*

3 Resolution of singularities

We demonstrate in this section how to resolve the singularities of a reflexive saturated parabolic sheaf over a smooth complex analytic manifold by blowing up successively along smooth centers. In the case of the process, we will get a locally abelian parabolic bundle whose parabolic degree will be preserved under pullback.

The following theorem [13, Theorem 4.11] plays a key role.

Theorem 3.1 *Let $\mathcal{F}$ be a coherent analytic sheaf over a complex analytic manifold X . There exists a bimeromorphic morphism $\pi : \tilde{X} \rightarrow X$, which is obtained by successively blowing*

up along smooth centers, such that the pullback of $\mathcal{F}$ modulo torsion is locally free and the torsion subsheaf is supported on the exceptional divisor.

The following corollary follows immediately.

Corollary 3.2 *Let $\mathcal{F}$ be a subsheaf of a locally free sheaf $\mathcal{E}$, then there exists a bimeromorphic morphism $\pi : \tilde{X} \rightarrow X$ such that the saturation of $\pi^*(\mathcal{F})$ in $\pi^*(\mathcal{E})$ is a subbundle of $\pi^*(\mathcal{E})$.*

Let $\mathcal{F}_*$ be a saturated reflexive parabolic sheaf over a pair (X, D) of a smooth complex analytic manifold X and a simple normal crossing divisor D . By Theorem 3.1, we resolve the underlying sheaf $\mathcal{F}$ to be a vector bundle $E \rightarrow \tilde{X}$ such that $\pi_1 : \tilde{X} \rightarrow X$ is some modification of X mentioned in Theorem 3.1. To save notation, if we blow up again, we will still denote the modification as $\pi : \tilde{X} \rightarrow X$ and the pullback of E as E . We also follow the convention that we always modulo the torsion of pullback of a sheaf and use the same notation for the pullbacked sheaf. Moreover, we will not change the notation for the inverse images of the irreducible components of the exceptional divisors generated on the process of the successive blow-ups. The proper transformation of D (D_i) will always be denoted as D^* (D_i^*) and the exceptional divisor will be denoted as $\mathcal{E}$. Without loss of generality, we always assume D^* and $\mathcal{E}$ intersects transversely.

Next, in order to obtain the locally abelian condition, we blow up to make sure at most two proper transforms of the irreducible components of the divisor intersect at a time. Restricting to each D_i^* , we have a filtration of subsheaves ${}^i\mathcal{F}_{a|D_i^*}$ on $E|_{D_i^*}$. Then we can apply Corollary 3.2 to get a filtration of subbundles on $E|_{D_i^*}$. Similarly, we can also make ${}^i\mathcal{F}_{a|D_{ij}^*} \cap {}^j\mathcal{F}_{b|D_{ij}^*}$ to be a subbundle of $E|_{D_{ij}^*}$ for any $a, b \in [0, 1]$ by blowing up again. Hence we have obtained a compatible parabolic structure on E over D^* . We denote this locally abelian parabolic bundle by E_* .

It follows easily from Lemma 2.4, Definition 1 that $\text{ch}_k(E_*) \cdot \pi^*\omega = \text{ch}_k(\mathcal{F}_*) \cdot \omega$ for $k = 1, 2$ and any Kähler class ω on X .

Hence we have:

Proposition 3.3 *Let $\mathcal{F}_*$ be a saturated reflexive parabolic sheaf over a pair (X, D) of a smooth complex analytic manifold X and a simple normal crossing divisor D . There exists a bimeromorphic morphism $\pi : \tilde{X} \rightarrow X$, which is obtained by successively blowing up along smooth centers, such that the torsion free quotient E of the pullback sheaf $\pi^*(\mathcal{F})$ admits a locally abelian structure over D^* . Moreover, we have $\text{ch}_k(E_*) \cdot \pi^*\omega = \text{ch}_k(\mathcal{F}_*) \cdot \omega$ for $k = 1, 2$ and any Kähler class ω on X .*

4 Constructions and analyses of metrics

In this section, we first construct a family of conical Kähler metrics ω_δ on $X \setminus D$ and a family of conical Kähler metrics $\omega_{\epsilon\delta}$ on $\tilde{X} \setminus D$. Then we construct a Hermitian metric $\hat{H}$ on the regular part of $\mathcal{F}_*$ which is compatible with the parabolic structure in the sense of Definition 1.1. The metrics will be used in Section 6 when we study the H-Y-M flow.

Let h_i be a Hermitian metric of $\mathcal{O}_X(D_i)$. Let σ_i denote the canonical section of $\mathcal{O}_X(D_i)$. The norm of σ_i with respect to h_i is denoted by $|\sigma_i|$. The canonical section of $\mathcal{O}_X(D)$ is denoted by σ and the norm of σ is denoted by $|\sigma|$. We assume that $|\sigma| < 1$. Also as in the last section, we don't change the notation for geometric objects under the pullback of $\pi : \tilde{X} \rightarrow X$.

4.1 Conical Kähler metrics

We construct a family of regularized metrics on X whose limit gives us a conical Kähler metric ω_δ on $X \setminus D$. This kind of regularization trick is commonly used to deal with the geometric problems on conical Kähler manifolds, cf. [14]. We introduce for any $0 \leq \nu$ and $0 < \delta < \frac{1}{2}$ the function

$$\chi_{\delta\nu}(t) := \delta \int_0^t \frac{(v^2 + s)^\delta - v^{2\delta}}{s} ds.$$

We define the following, for some positive number $C > 0$,

$$\omega_{\delta\nu} := \omega + C \cdot \sum_{i \in I} \sqrt{-1} \cdot \partial \bar{\partial} \chi_{\delta\nu}(|\sigma_i|^2).$$

Proposition 4.1 *For C sufficiently small, if $0 < \nu$, $\omega_{\delta\nu}$ is a Kähler metric on X for any δ , and if $\nu = 0$, $\omega_\delta := \omega_{\delta 0}$ is a Kähler metric on $X \setminus D$ for any δ .*

Proof Direct calculations show that

$$\begin{aligned} \sqrt{-1} \cdot \partial \bar{\partial} \chi_{\delta\nu}(|\sigma_i|^2) &= \sqrt{-1} \cdot \delta^2 \cdot (v^2 + |\sigma_i|^2)^\delta \cdot \\ &\quad \frac{\langle \partial_{h_i} \sigma_i, \partial_{h_i} \sigma_i \rangle}{v^2 + |\sigma_i|^2} - \sqrt{-1} \cdot \delta \cdot ((v^2 + |\sigma_i|^2)^\delta - v^{2\delta}) \cdot F_{h_i}. \end{aligned}$$

The proposition follows from the uniform boundedness of the quantities: $\frac{\langle \partial_{h_i} \sigma_i, \partial_{h_i} \sigma_i \rangle}{v^2 + |\sigma_i|^2}$, F_{h_i} , $\delta \cdot (v^2 + |\sigma_i|^2)^\delta$ and $\delta^2 \cdot (v^2 + |\sigma_i|^2)^\delta$. $\square$

The above equation also implies that $\omega_{\delta\nu}$ converges in C_{loc}^∞ -topology to ω_δ as $\nu \rightarrow 0$. The Kähler metrics ω_δ behave well around any point of D in the following sense, which is clear by construction.

Lemma 4.2 *Let $x \in D_J$ with $J \subset I$, we may choose a small coordinate neighborhood centered at x such that the defining function of D is $z_1 \cdots z_k = 0$. Then there exists a positive constants C_1 such that*

$$C_1^{-1} \cdot \omega_\delta \leq \sqrt{-1} \cdot \delta^2 \cdot \sum_{i=1}^k \frac{dz_i \wedge d\bar{z}_i}{|z_i|^{2-2\delta}} + \sqrt{-1} \cdot \partial \bar{\partial} |z|^2 \leq C_1 \cdot \omega_\delta.$$

Notice that the pullback of ω_δ by π is a degenerated conical Kähler metric on $\tilde{X} \setminus D$. Fix a normalized Kähler metric $\omega_{\tilde{X}}$ on $\tilde{X}$ such that $\int_{\tilde{X}} \omega_{\tilde{X}}^n = 1$. We put $\omega_{\epsilon\delta} := \omega_\delta + \epsilon \cdot \omega_{\tilde{X}}$. Then $\omega_{\epsilon\delta}$ is a conical Kähler metric of $\tilde{X} \setminus D$. And we have the following proposition.

Proposition 4.3 *The Kähler manifold $(\tilde{X} \setminus D, \omega_{\epsilon\delta})$ satisfies the following three assumptions:*

1. *The volume of $\tilde{X} \setminus D$ is uniformly bounded independent of ϵ and δ .*
2. *There exists an exhaustion function ϕ with $\sqrt{-1} \Lambda_{\omega_{\epsilon\delta}} \partial \bar{\partial} \phi$ bounded.*
3. *If f is a bounded positive function on $\tilde{X} \setminus D$ such that $\Delta_{\epsilon\delta} f \leq B$ for some positive function $B \in L^p$ ($p > n$), then $\|f\|_{L^\infty} \leq C_2(\|B\|_{L^p} + \|f\|_{L^1})$. The constant C_2 is independent of ϵ and δ .*

Here $\Delta_{\epsilon\delta} = 2\sqrt{-1} \Lambda_{\omega_{\epsilon\delta}} \partial \bar{\partial}$ is the negative Laplace operator and the Laplace operators appear in the rest of the paper will always be the negative one.

Proof The first assertion follows directly from the previous lemma. We put $\phi := \log |\sigma|$. Then the second assertion follows from the Poincaré-Lelong formula.

To prove the third assertion, we need a very important result of [15] which will also be used later. Let (X, ω_X) be a compact Kähler manifold with a normalized Kähler metric ω_X such that $\int_X \omega_X^n = 1$. Suppose that the complex dimension of X is n . Given any Kähler metric ω on X , we denote its volume by $V_\omega := [\omega]^n$ and define the relative volume density by

$$e^{\lambda_\omega} := \frac{1}{V_\omega} \frac{\omega^n}{\omega_X^n}.$$

Given $p \geq 1$ we define the p -th Nash-Yau entropy by

$$\mathcal{N}_p(\omega) := \frac{1}{V_\omega} \int_X \left| \log \frac{1}{V_\omega} \cdot \frac{\omega^n}{\omega_X^n} \right|^p \cdot \omega^n.$$

For a given nonnegative continuous function $\gamma \in C^0(X)$, and given parameters $0 < A \leq \infty$, $K > 0$, we consider the following subset of the space of Kähler metrics on X :

$$\mathcal{W} := \mathcal{W}(n, p, A, K, \gamma) := \left\{ \omega : [\omega] \cdot [\omega_X]^{n-1} < A, \mathcal{N}_p \leq K, e^{\lambda_\omega} \geq \gamma \right\}.$$

The following theorem concerning the Sobolev inequality with respect to the Kähler metrics in $\mathcal{W}$ was obtained in [15, Theorem 2.1].

Theorem 4.4 (Uniform Sobolev Inequality) *Given $p > n$ and the zero locus of γ has Hausdorff dimension less than $2n - 1$, there exist $q = q(n, p) > 1$ and $C_3 = C_3(n, p, A, K, \gamma, q) > 0$ such that for any $\omega \in \mathcal{W}$ and any $u \in L^2_1(X)$, we have the following Sobolev-type inequality*

$$\left(\frac{1}{V_\omega} \int_X |u - \bar{u}|^{2q} \omega^n \right)^{\frac{1}{q}} \leq C_3 \frac{1}{V_\omega} \int_X |\nabla u|^2 \omega^n,$$

where $\bar{u} := \frac{1}{V_\omega} \int_X u \omega^n$.

Now we return to the proof of the third assertion. We will apply the above theorem to investigate a family of metrics on $\tilde{X}$. We put

$$\omega_{\epsilon\delta\nu} := \omega_{\delta\nu} + \epsilon \cdot \omega_{\tilde{X}}.$$

This is a smooth family of Kähler metrics on $\tilde{X}$. Since $\omega_{0\delta\nu}$ only degenerates on the exceptional divisor $\mathcal{E}$, $\omega_{\epsilon\delta\nu}$ is supported by a continuous function γ with vanishing locus of Hausdorff dimension $2n - 2$. In order to apply the above theorem, we only need to check the Nash-Yau entropy condition. Recall that $\mathcal{E} \cup D$ is simple normal crossing. Hence we may choose a local coordinate patch $z = (z', z'', z''')$, such that $D = \bigcup_{i=1}^{k'} \{z'_i = 0\}$ and $\mathcal{E} = \bigcup_{j=1}^{k''} \{z''_j = 0\}$. We may assume that $|z| < 1$. We put

$$f_{\epsilon\delta\nu} := \frac{\omega_{\epsilon\delta\nu}^n}{\omega_{\tilde{X}}^n}.$$

Then $f_{\epsilon\delta\nu}$ either degenerates to 0 on $\mathcal{E}$ with a speed $\prod_j |z''_j|^a$ or blows up to infinity on D with a speed $\prod_i |z'_i|^{-a}$ with some uniform $a > 0$. Therefore, if we fix a $p > n$, we have

$$|\log(f_{\epsilon\delta\nu})|^p \leq \frac{C_{15}}{\delta^{k'} \cdot \prod_{i=1}^{k'} |z'_i|^{\frac{\delta}{2}} \cdot \prod_{j=1}^{k''} |z''_j|^{\frac{1}{2}}}$$

with C_{15} a constant independent of δ , ϵ and ν . Then it follows from Lemma 4.2 that

$$|\log(f_{\epsilon\delta\nu})|^p \cdot \omega_{\epsilon\delta\nu}^n \leq \frac{C_{15} \cdot \delta^{k'}}{\prod_{i=1}^{k'} |z'_i|^{2-\frac{3\delta}{2}} \cdot \prod_{j=1}^{k''} |z''_j|^{\frac{1}{2}}}.$$

Hence the integral of $|\log(f_{\epsilon\delta\nu})|^p \cdot \omega_{\epsilon\delta\nu}^n$ is uniformly bounded. Consequently, we can find suitable constants $p > n$, A and K such that the family of metrics $\omega_{\epsilon\delta\nu}$ belongs to $\mathcal{W}(n, p, A, K, \gamma)$. Then the above theorem implies the following lemma.

Lemma 4.5 *For any $f \in C_0^\infty(\tilde{X} \setminus D^*)$ and with respect to any conical Kähler metric $\omega_{\epsilon\delta}$ on $\tilde{X} \setminus D^*$, we have the uniform Sobolev inequality*

$$\|f\|_{L^q} \leq C_4 \|f\|_{L^2_1}$$

with C_4 independent of ϵ and δ .

Proof This is true because f is compactly supported in $\tilde{X} \setminus D^*$ and $\omega_{\epsilon\delta\nu}$ converges in C_{loc}^∞ -topology to $\omega_{\epsilon\delta}$ as ν tends to 0. $\square$

Now we are ready to prove the third assertion by Moser's iteration. Let f be a positive bounded function which satisfies the assumption of the third assertion, i.e. $\Delta_{\omega_{\epsilon\delta}} f = \sqrt{-1} \Lambda_{\omega_{\epsilon\delta}} \partial \bar{\partial} f \leq B$. We put $f_\nu := (f + \nu \log |\sigma|)^+$ with $0 < \nu < 1$. Then we have

$$\Delta_{\omega_{\epsilon\delta}} f_\nu \leq B_\nu$$

in the weak sense, where B_ν converges to B in C^∞ -topology as ν tends to 0. As f_ν lies in the Sobolev closure of $C_0^\infty(\tilde{X} \setminus D^*)$, hence we may apply Moser's iteration to obtain the estimate

$$\|f_\nu\|_{L^\infty} \leq C_5 (\|f_\nu\|_{L^1} + \|B_\nu\|_{L^p})$$

with $p > n$. Taking the limit as ν tends to 0 completes the proof. $\square$

The uniform Sobolev inequality also implies the following uniform upper bound for the heat kernels on $\tilde{X} \setminus D$ with respect to the family of conical metrics $\omega_{\epsilon\delta}$.

Proposition 4.6 *Let $K_{\epsilon\delta}$ be the heat kernel with respect to $\omega_{\epsilon\delta}$, then for any $\tau > 0$, there exists a constant $C_K(\tau)$ which is independent of ϵ and δ such that*

$$0 < K_{\epsilon\delta}(x, y, t) \leq C_K(\tau) \left(t^n \exp\left(-\frac{d_{\epsilon\delta}(x, y)}{(4+\tau)t}\right) + 1 \right)$$

where $d_{\epsilon\delta}(\cdot, \cdot)$ is the distance function of $\tilde{X} \setminus D^*$ with respect to the metric $\omega_{\epsilon\delta}$.

Proof Recall that $\omega_{\epsilon\delta\nu}$ is a family of smooth metrics on the compact manifold $\tilde{X}$ such that the uniform Sobolev inequality is satisfied. On the one hand, by combining the diagonal heat kernel estimate shown in [8] and the Gaussian upper bound estimate in [12, Theorem 1.1], we have

$$0 < K_{\epsilon\delta\nu}(x, y, t) \leq C_K(\tau) \left(t^n \exp\left(-\frac{d_{\epsilon\delta\nu}(x, y)}{(4+\tau)t}\right) + 1 \right).$$

On the other hand, it is easy to show that $K_{\epsilon\delta\nu}$ converges uniformly to the heat kernel $K_{\epsilon\delta}$ on compact subsets of $(\tilde{X} \setminus D^*) \times (\tilde{X} \setminus D^*)$ as ν tends to 0. Hence the proposition follows. $\square$

4.2 Parabolic metric

The target of this subsection is to construct a Hermitian metric $\widehat{H}$ on the regular part of $\mathcal{F}_*$ which is compatible with the parabolic structure in the sense of Definition 1.1. We also wish it to satisfy the following equations in the sense of currents on X :

$$\begin{aligned}\mathrm{ch}_1(\mathcal{F}_*) &= \frac{\sqrt{-1}}{2\pi} \mathrm{Tr}(F_{\widehat{H}}) \\ \mathrm{ch}_1(\mathcal{F}_*)^2 &= \left(\frac{\sqrt{-1}}{2\pi} \right)^2 \mathrm{Tr}(F_{\widehat{H}}) \wedge \mathrm{Tr}(F_{\widehat{H}}) \\ \mathrm{ch}_2(\mathcal{F}_*) &= \frac{1}{2} \left(\frac{\sqrt{-1}}{2\pi} \right)^2 \mathrm{Tr}(F_{\widehat{H}} \wedge F_{\widehat{H}}).\end{aligned}$$

Definition 4.1 If a Hermitian metric H defined on $\mathcal{F}_*|_{X^\circ \setminus D}$ satisfies the first equation above, we call it adapted to $\mathcal{F}_*$ in codimension 1. If it satisfies all of the equations, we call it adapted to $\mathcal{F}_*$ in codimension 2.

In view of Proposition 3.3, it suffices to construct a Hermitian metric H_0 on the regular part of the locally abelian parabolic bundle $E'_* := E \otimes [-P_0]$ which is adapted in codimension 2. Indeed, should this be completed, we may choose a Hermitian metric h_{P_0} for the line bundle $\mathcal{O}_{\widetilde{X}}(P_0)$ and then $\widehat{H} := H_0 \otimes h_{P_0}$ will be a Hermitian metric on $E|_{\widetilde{X} \setminus D^*}$ which in turn will induce a metric on the regular part of $\mathcal{F}_*$. Since $E \otimes [-P] \cong \mathcal{F}$, it is easy to see that $\widehat{H}$ is adapted to $\mathcal{F}_*$ in codimension 2 because the singular locus has codimension at least 3.

We introduce an approach to construct an adapted metric on a locally abelian parabolic bundle. The notation used in the construction process will be independent of that used elsewhere. Firstly, it is important to have a smooth decomposition of the bundle in a small neighborhood of the divisor D which is in some sense compatible with the parabolic structure.

4.2.1 Smooth decomposition

Local patching Let X be a complex manifold. Let $D = \bigcup_{1 \leq i \leq \ell} D_i$ be a simple normal crossing divisor. We set $Y = \bigcap_{i=1}^{\ell} D_i$. We set $S = \{0 \leq a < 1\}$. For any $1 \leq i \leq \ell$, let $q_i : S^\ell \rightarrow S$ denote the projection onto the i -th component.

Let $F_* = (F, \mathbf{F})$ be a locally abelian parabolic bundle on X . Recall that $\mathbf{F} = ({}^iF \mid i = 1, \dots, \ell)$ is a tuple of decreasing filtrations iF of $F|_{D_i}$ by holomorphic subbundles indexed by S such that the following holds.

- For any $1 \leq i \leq \ell$ and $a \in S$, there exists $\epsilon > 0$ such that ${}^iF_a = {}^iF_{a-\epsilon}$.
- For any $P \in Y$, there exist a neighbourhood $\mathcal{U}_P$ of P in X and a holomorphic decomposition $F|_{\mathcal{U}_P} = \bigoplus_{a \in S^\ell} {}^P G_a$ such that

$$\bigoplus_{q_i(a) \geq @} {}^P G_a|_{D_i \cap \mathcal{U}_P} = {}^iF|_{D_i \cap \mathcal{U}_P}.$$

Here the symbol “@” denotes a real variable taking values from $[0, 1[$ and we regard

$$\bigoplus_{q_i(a) \geq @} {}^P G_a|_{D_i \cap \mathcal{U}_P}$$

as a left continuous filtration.

Let $X_2 \subset X_1 \subset X$ be open subsets such that the closure of X_2 is contained in X_1 . Suppose that there exists a C^∞ -decomposition

$$F|_{X_1} = \bigoplus_{a \in S^\ell} {}^1G_a$$

such that the following holds:

$$\bigoplus_{q_i(a) \geq @} {}^1G_{a|D_i \cap X_1} = {}^iF|_{D_i \cap X_1}.$$

Proposition 4.7 *There exist a neighbourhood $\mathcal{U}$ of Y and a C^∞ -decomposition*

$$F|_{\mathcal{U}} = \bigoplus_{a \in S^\ell} G_a$$

such that the following holds:

- $\bigoplus_{q_i(a) \geq @} G_{a|D_i \cap \mathcal{U}} = {}^iF|_{D_i \cap \mathcal{U}}$.
- $G_{a|X_2} = {}^1G_{a|X_2}$.

Proof Since locally, F_* is a direct sum of parabolic line bundles, there exists a tuple of open subsets $\mathcal{U}_k$ ($k \in \Gamma$) of X such that the following holds:

- $Y \subset \mathcal{V} := \bigcup \mathcal{U}_k \cup X_1$
- On each $\mathcal{U}_k$, there exists a C^∞ -decomposition

$$F|_{\mathcal{U}_k} = \bigoplus_{a \in S^\ell} {}^kG_a$$

such that the following holds:

$$\bigoplus_{q_i(a) \geq @} {}^kG_{a|D_i \cap \mathcal{U}_k} = {}^iF|_{D_i \cap \mathcal{U}_k}.$$

Let $\{\chi_k\} \cup \{\chi_{X_1}\}$ be a partition of unity on $\mathcal{V}$ subordinate to the open covering $\mathcal{V} = \bigcup \mathcal{U}_k \cup X_1$. We may assume that $\chi_{X_1} = 1$ on X_2 . Let $\psi : S^\ell \rightarrow \mathbb{C}$ be an injection. We set

$$f_k = \sum_{a \in S^\ell} \psi(a) \cdot \text{id}_{{}^kG_a}$$

and

$$f_{X_1} = \sum_{a \in S^\ell} \psi(a) \cdot \text{id}_{{}^1G_a}.$$

We obtain the following C^∞ -endomorphism of F on $\mathcal{V}$:

$$f = \sum_{k \in \Gamma} \chi_k \cdot f_k + \chi_{X_1} \cdot f_{X_1}.$$

In the mean time, we obtain the following lemma by the construction.

Lemma 4.8 *The restriction $f|_Y$ preserves the filtrations iF ($i = 1, \dots, \ell$) of $F|_Y$. Moreover, the induced endomorphism on ${}^\ell\text{Gr}_a(F|_Y)$ equals the multiplication of $\psi(a)$.*

As a result, we obtain the eigen decomposition

$$(F, f)|_Y = \bigoplus_{\alpha \in \mathbb{C}} (K_\alpha, \alpha \cdot \text{id}_{K_\alpha}).$$

There exists a neighbourhood $\mathcal{U}$ of Y and a decomposition

$$(F, f)|_{\mathcal{U}} = \bigoplus_{\alpha \in \mathbb{C}} (K_{\mathcal{U}, \alpha}, f_\alpha)$$

such that $K_{\mathcal{U}, \alpha}|_Y = K_\alpha$. If $\mathcal{U}$ is sufficiently small, any eigenvalues of $f_\alpha|_P$ ($P \in \mathcal{U}$) are close to α . In particular, we may assume that there are no common eigenvalues of f_α and f_β ($\alpha \neq \beta$). We set

$$G_a = K_{\mathcal{U}, \psi(a)}.$$

The restriction $f|_{D_i \cap \mathcal{U}}$ preserves the filtration ${}^i F$. We have

$${}^i F_b|_Y = \bigoplus_{q_i(a) \geq b} K_{\psi(a)}.$$

Then, we obtain $\bigoplus_{q_i(a) \geq b} G_a|_{D_i \cap \mathcal{U}} = {}^i F|_{D_i \cap \mathcal{U}}$. Because $\chi_{X_1} = 1$ on X_2 , we have $G_a|_{X_2} = {}^1 G_a|_{X_2}$. $\square$

Glocal decomposition With the above proposition at hand it is easy to get a global C^∞ -decomposition near D . Let X be a complex manifold. Let $D = \bigcup_{i \in \Lambda} D_i$ be a simple normal crossing divisor. For any $I \subset \Lambda$, we set $D_I = \bigcap_{i \in I} D_i$, $\partial D_I = \bigcup_{j \in \Lambda \setminus I} (D_I \cap D_j)$ and $D_I^\circ = D_I \setminus \partial D_I$. For any $I \subset J \subset \Lambda$, let $q_{I,J} : S^J \rightarrow S^I$ denote the projection. As before, let $F_* = (F, F)$ be a parabolic bundle on X . Let $F = ({}^i F | i \in \Lambda)$ be the parabolic structure.

Proposition 4.9 *There exist neighbourhoods $\mathcal{U}_i$ ($i \in \Lambda$) of D_i and C^∞ -decompositions*

$$F|_{\mathcal{U}_i} = \bigoplus_{a \in S} {}^i G_a$$

such that the following holds:

- $\bigoplus_{a \geq b} {}^i G_a|_{D_i} = {}^i F_b$ holds for any $b \in S$.
- For any $I \subset \Lambda$ and $a = ({}^i a | i \in I) \in S^I$, on $\mathcal{U}_I = \bigcap_{i \in I} \mathcal{U}_i$, we set ${}^I G_a = \bigcap_{i \in I} {}^i G_{i a}|_{\mathcal{U}_i}$. Then, $F|_{\mathcal{U}_I} = \bigoplus_{a \in S^I} {}^I G_a$ holds.

Proof By using a descending induction on $|I|$, we shall construct neighbourhoods $\mathcal{V}_I$ ($I \subset \Lambda$) of D_I and decompositions

$$F|_{\mathcal{V}_I} = \bigoplus_{a \in S^I} {}^I G_a$$

such that the following holds:

- $\bigoplus_{q_{i,I}(a) \geq b} {}^I G_a|_{D_i \cap \mathcal{V}_I} = {}^i F_b|_{D_i \cap \mathcal{V}_I}$ holds for any $b \in S$.
- For $I \subset J$ and $a \in S^I$, we have ${}^I G_a|_{\mathcal{V}_J} = \bigoplus_{q_{I,J}(b)=a} {}^J G_b$.

Suppose that we have already constructed such decompositions for $J \subset \Lambda$ with $|J| \geq k_0 + 1$. Let $I \subset \Lambda$ with $|I| = k_0$. For $I \subset J \subset \Lambda$ with $\mathcal{U}_J \neq \emptyset$ and for $a \in S^I$, we set

$${}^J G_a := \bigoplus_{q_{I,J}(b)=a} {}^J G_b.$$

We obtain the following decomposition:

$$F|_{\mathcal{U}_J} = \bigoplus_{a \in S^I} {}^J G_a$$

By the condition (b), we obtain ${}^J G_a|_{\mathcal{U}_K} = {}^K G_a$ for any $J \subset K$ and $a \in S^I$. From ${}^J G_a$ ($I \subsetneq J$), we obtain a C^∞ -subbundle ${}^{1,I} G_a$ of F on $\mathcal{U}_{I,1} = \bigcup_{I \subsetneq J} \mathcal{U}_J$. We obtain the decomposition

$$F|_{\mathcal{U}_{I,1}} = \bigoplus_{a \in S^I} {}^{1,I} G_a$$

Let $\mathcal{U}_{I,2} \subset \mathcal{U}_{I,1}$ be an open neighbourhood of ∂D_I whose closure is contained in $\mathcal{U}_{I,1}$. By Proposition 4.7, there exist a neighbourhood $\mathcal{V}_I$ of D_I and a decomposition

$$F|_{\mathcal{U}_I} = \bigoplus_{a \in S^I} {}^I G_a$$

such that the condition (a) holds for I and that ${}^I G_a|_{\mathcal{U}_{I,2}} = {}^{1,I} G_a|_{\mathcal{U}_{I,2}}$. We replace $\mathcal{V}_J$ with $\mathcal{V}_J \cap \mathcal{U}_I$ for any $J \ni I$. Then, we obtain the claim of the proposition. $\square$

4.2.2 Construction of metric

Let H_1 be a Hermitian metric of F such that the following holds:

- The decomposition $F|_{\mathcal{U}_I} = \bigoplus {}^I G_a$ is orthogonal with respect to $H_1|_{\mathcal{U}_I}$.

Let $h^{(i)}$ ($i \in \Lambda$) be Hermitian metrics of $\mathcal{O}_X(D_i)$. We set $\tau_i = h^{(i)}(1, 1)$. We may assume that τ_i is constantly 1 on $X \setminus \mathcal{U}_i$. Let $\tau := \bigotimes_{i \in \Lambda} \tau_i$ be the induced square length of the canonical section of D . Let s_i be the automorphism of $F|_{X \setminus D_i}$ such that the following holds:

- $s_i = \text{id}$ on $X \setminus \mathcal{U}_i$.
- $s_i = \bigoplus_{a \in S} \tau_i^{-a} \text{id}_{G_a}$ on $\mathcal{U}_i$.

We obtain the automorphism $s = \prod_{i \in \Lambda} s_i$ of F which is self-adjoint with respect to H_1 . We define the C^∞ -Hermitian metric H_0 of $F|_{X \setminus D}$ by $H_0(u, v) = H_1(su, v)$ for any local sections u, v of F . And we may extend H_0 smoothly to X .

4.2.3 Adaptedness

It follows from the construction that for any $[\eta] \in H^{n-1, n-1}(X, \mathbb{C})$ we have:

Proposition 4.10 $\text{ch}_1(F_*) \cdot [\eta] = \frac{\sqrt{-1}}{2\pi} \int_{X \setminus D} \text{Tr}(F_{\nabla_{H_0}}) \wedge \eta$.

Let ε be the largest gap of the weights and κ be the smallest gap of the weights.

Let $(U, z_1, \dots, z_n)$ be a holomorphic coordinate neighbourhood of X such that $D \cap U = \bigcup_{i=1}^\ell \{z_i = 0\}$. For $1 \leq i \leq \ell$, we have $\lambda(i) \in \Lambda$ such that $D_{U, \lambda(i)} := D_{\lambda(i)} \cap U = \{z_i = 0\}$. We set $I = \{\lambda(i) \mid i = 1, \dots, \ell\}$. By shrinking U , we assume $U \subset \mathcal{U}_I$. We obtain the induced C^∞ -decomposition

$$F|_U = \bigoplus_{a \in S^I} {}^I G_a.$$

For any $1 \leq k \leq \ell$ and $a \in S$, we set

$${}^k G_a = \bigoplus_{q_{\lambda(k)}(a)=a} {}^I G_a.$$

We obtain the decomposition

$$F|_U = \bigoplus_{a \in S} {}^k G_a. \quad (4)$$

Let ι_a be the injection from ${}^I G_a$ to F and π_a be the orthogonal projection from F to ${}^I G_a$ with respect to the metric H_1 . We set $\nabla_a := \pi_a \cdot \nabla_{H_1} \cdot \iota_a$ as the induced connection on ${}^I G_a$. Then $\tilde{\nabla} := \bigoplus_a \nabla_a$ is a unitary connection which preserves the decomposition. We have for any a and k , restricting on ${}^k G_{a|D_k} = {}^k \text{Gr}_a F_*$, $\tilde{\nabla} = \nabla_{H_1}$.

We consider the decomposition

$$\nabla_{H_1} = \tilde{\nabla} + \Psi$$

where Ψ is a section of

$$\bigoplus_{a \neq b} A^1(\text{Hom}({}^I G_b, {}^I G_a)).$$

Let $\Psi = \Psi^{1,0} + \Psi^{0,1}$ be the decomposition into the $(1, 0)$ -part and the $(0, 1)$ -part. We express

$$\Psi^{1,0} = \sum \Psi_j^{1,0} dz_j, \quad \Psi^{0,1} = \sum \Psi_j^{0,1} d\bar{z}_j.$$

We have the decompositions

$$\Psi_j^{p,q} = \sum (\Psi_j^{p,q})_{a,b}, \quad (\Psi_j^{p,q})_{a,b} \in A^0(\text{Hom}({}^I G_b, {}^I G_a)).$$

By the construction, $(\Psi_j^{p,q})_{a,a} = 0$ for any $1 \leq j \leq n$ and $a \in S^I$.

We denote F_{∇} as the curvature tensor with respect to some connection ∇ . Then we have the following lemma.

Lemma 4.11 $|F_{\nabla_{H_0}}|_{H_0} = O(\tau^{-\frac{1+\varepsilon}{2}})$

Proof It suffices to consider in U . We have

$$F_{\nabla_{H_1}} = F_{\tilde{\nabla}} + \tilde{\nabla}(\Psi) + \Psi \wedge \Psi.$$

Hence

$$\begin{aligned} F_{\nabla_{H_0}} &= F_{\nabla_{H_1}} + \bar{\partial}(s^{-1} \nabla_{H_1}^{1,0} s) \\ &= F_{\tilde{\nabla}} + \tilde{\nabla}(\Psi) + \Psi \wedge \Psi + \bar{\partial}(s^{-1} \tilde{\nabla}^{1,0} s + s^{-1} [\Psi, s]) \\ &= B + \bar{\partial}(s^{-1} \Psi s) \end{aligned}$$

where B is smooth. We put $A := s^{-1} \Psi s$. Then we have $(\bar{\partial} A)_{a,b} = O(|\tau_I|^{b-a-\frac{1}{2}})$ where $\frac{1}{2} \in S^I$. Since H_0 is diagonal with respect to the decomposition, it suffices to estimate

$$\left((\bar{\partial} A)_{a,b} \cdot (\bar{\partial} A)_{a,b} \cdot H_{2,a,a} \cdot H_{2,b,b}^{-1} \right)^{\frac{1}{2}}.$$

Simple calculation shows that

$$\left((\bar{\partial} A)_{a,b} \cdot (\bar{\partial} A)_{a,b} \cdot H_{2,a,a} \cdot H_{2,b,b}^{-1} \right)^{\frac{1}{2}} = O(\tau_I^{\frac{b-a-1}{2}}) = O(\tau^{-\frac{1+\varepsilon}{2}}).$$

□

Next we show the adaptedness in codimension 2.

We still consider in U but fix a $1 \leq k \leq \ell$. We consider the decomposition

$$\nabla_{H_1} = \tilde{\nabla} + \Psi,$$

as above.

Let $\Psi = \Psi^{1,0} + \Psi^{0,1}$ be the decomposition into the $(1, 0)$ -part and the $(0, 1)$ -part. We express

$$\Psi^{1,0} = \sum \Psi_j^{1,0} dz_j, \quad \Psi^{0,1} = \sum \Psi_j^{0,1} d\bar{z}_j.$$

We have the decompositions

$$\Psi_j^{p,q} = \sum (\Psi_j^{p,q})_{a,b}, \quad (\Psi_j^{p,q})_{a,b} \in A^0(\text{Hom}(G_b, G_a)).$$

By the construction, $(\Psi_j^{p,q})_{a,a} = 0$ for any $1 \leq j \leq n$ and $a \in S$.

Lemma 4.12 *For $j \neq k$ and $a < b$, we have*

$$(\Psi_j^{0,1})_{a,b} = O(|z_k|).$$

Proof Note that kF is a left continuous decreasing filtration of $F|_{D_k}$ by holomorphic subbundles. Hence for any $b \in S$,

$$(\Psi_j^{0,1})|_{D_k \cap U}({}^kF_b) \subset {}^kF_{\geq b}.$$

It implies the claim of the lemma. $\square$

Because $\Psi^{0,1} = -(\Psi^{1,0})_{H_1}^\dagger$, we obtain the following for $j \neq k$ and $a > b$:

$$(\Psi_j^{1,0})_{a,b} = O(|z_k|).$$

As a result, we obtain the following lemma.

Lemma 4.13 *If $a > b$ and $j \neq k$, then*

$$(s^{-1}\Psi_j^{1,0}s)_{a,b} = O\left(|z_k|^{1-2\varepsilon} \prod_{i \neq k} |z_i|^{-2\varepsilon}\right).$$

If $a < b$, then for any j ,

$$(s^{-1}\Psi_j^{1,0}s)_{a,b} = O\left(\prod_{1 \leq i \leq \ell} |z_i|^{-2\varepsilon}\right).$$

Let $F_{\nabla_{H_1}}$ denote the curvature of ∇_{H_1} . Let $F_{\tilde{\nabla}}$ denote the curvature of $\tilde{\nabla}$. We obtain

$$F_{\nabla_{H_1}} = F_{\tilde{\nabla}} + \tilde{\nabla}(\Psi) + \Psi \wedge \Psi.$$

Let $\tilde{\nabla} = \tilde{\nabla}^{1,0} + \tilde{\nabla}^{0,1}$ denote the decomposition into the $(1, 0)$ -part and the $(0, 1)$ -part. We have

$$s^{-1}\nabla_{H_1}^{1,0}(s) = s^{-1}\tilde{\nabla}^{1,0}s + s^{-1}[\Psi^{1,0}, s].$$

We also obtain

$$\begin{aligned} \bar{\partial}(s^{-1}\nabla_{H_1}^{1,0}s) &= \tilde{\nabla}^{0,1}(s^{-1}\tilde{\nabla}^{1,0}s) + \tilde{\nabla}^{0,1}(s^{-1}[\Psi^{1,0}, s]) \\ &\quad + [\Psi^{0,1}, s^{-1}\tilde{\nabla}^{1,0}s] + [\Psi^{0,1}, s^{-1}[\Psi^{1,0}, s]]. \end{aligned} \quad (5)$$

Because the trace is 0 for any section of $\text{Hom}(G_a, G_b)$ ($a \neq b$), we obtain

$$\begin{aligned} \text{Tr} \left(F_{\nabla_{H_1}} \cdot s^{-1} \nabla_{H_1}^{1,0} s \right) &= \text{Tr} \left(F_{\tilde{\nabla}} \cdot s^{-1} \tilde{\nabla}^{1,0} s + \tilde{\nabla}(\Psi) \cdot s^{-1} [\Psi^{1,0}, s] \right. \\ &\quad \left. + (\Psi \wedge \Psi) \cdot (s^{-1} \tilde{\nabla}^{1,0} s) + (\Psi \wedge \Psi) \cdot s^{-1} [\Psi^{1,0}, s] \right). \end{aligned} \quad (6)$$

We also obtain

$$\begin{aligned} \text{Tr} \left(\bar{\partial}_F (s^{-1} \nabla_{H_1}^{1,0} s) \cdot s^{-1} \nabla_{H_1}^{1,0} s \right) &= \text{Tr} \left(\tilde{\nabla}^{0,1} (s^{-1} \tilde{\nabla}^{1,0} s) \cdot s^{-1} \tilde{\nabla}^{1,0} s \right) \\ &\quad + \text{Tr} \left(\tilde{\nabla}^{0,1} (s^{-1} [\Psi^{1,0}, s]) \cdot s^{-1} [\Psi^{1,0}, s] \right) \\ &\quad + \text{Tr} \left([\Psi^{0,1}, s^{-1} \tilde{\nabla}^{1,0} s] \cdot s^{-1} [\Psi^{1,0}, s] \right) \\ &\quad + \text{Tr} \left([\Psi^{0,1}, s^{-1} [\Psi^{1,0}, s]] \cdot s^{-1} \tilde{\nabla}^{1,0} s \right) \\ &\quad + 2 \text{Tr} \left(\Psi^{0,1} \cdot (s^{-1} [\Psi^{1,0}, s])^2 \right). \end{aligned} \quad (7)$$

Note that

$$\begin{aligned} \text{Tr} \left(\tilde{\nabla}^{0,1} (s^{-1} [\Psi^{1,0}, s]) \cdot s^{-1} [\Psi^{1,0}, s] \right) &= \text{Tr} \left(s^{-1} [\tilde{\nabla}^{0,1} \Psi^{1,0}, s] \cdot s^{-1} [\Psi^{1,0}, s] \right) \\ &\quad - \text{Tr} \left([s^{-1} \Psi^{1,0} s, s^{-1} \tilde{\nabla}^{0,1} s] \cdot s^{-1} [\Psi^{1,0}, s] \right). \end{aligned} \quad (8)$$

$$\begin{aligned} \text{Tr} \left([\Psi^{0,1}, s^{-1} \tilde{\nabla}^{1,0} s] \cdot s^{-1} [\Psi^{1,0}, s] \right) &= -\text{Tr} \left(s^{-1} \tilde{\nabla}^{1,0} s \cdot [\Psi^{0,1}, \Psi^{1,0}] \right) \\ &\quad + \text{Tr} \left([\Psi^{0,1}, s^{-1} \Psi^{1,0} s] \cdot s^{-1} \tilde{\nabla}^{1,0} s \right). \end{aligned} \quad (9)$$

$$\begin{aligned} \text{Tr} \left([\Psi^{0,1}, s^{-1} [\Psi^{1,0}, s]] \cdot s^{-1} \tilde{\nabla}^{1,0} s \right) &= -\text{Tr} \left([\Psi^{0,1}, \Psi^{1,0}] \cdot s^{-1} \tilde{\nabla}^{1,0} s \right) \\ &\quad + \text{Tr} \left([\Psi^{0,1}, s^{-1} \Psi^{1,0} s] \cdot s^{-1} \tilde{\nabla}^{1,0} s \right). \end{aligned} \quad (10)$$

For any 2-form τ , let $\tau^{1,1}$ denote the $(1, 1)$ -part of τ . We obtain

$$\begin{aligned} &2 \text{Tr} \left(F_{\nabla_{H_1}} s^{-1} \nabla_{H_1}^{1,0} s \right) + \text{Tr} \left(\bar{\partial} (s^{-1} \nabla_{H_1}^{1,0} s) \nabla_{H_1}^{1,0} s \right) \\ &= \text{Tr} \left(2 F_{\tilde{\nabla}}^{1,1} s^{-1} \tilde{\nabla}^{1,0} s + \tilde{\nabla}^{0,1} (s^{-1} \tilde{\nabla}^{1,0} s) s^{-1} \tilde{\nabla}^{1,0} s \right) \\ &\quad + \text{Tr} \left(2 \tilde{\nabla}(\Psi)^{1,1} s^{-1} [\Psi^{1,0}, s] + 2 (\Psi \wedge \Psi)^{1,1} s^{-1} [\Psi^{1,0}, s] \right) \\ &\quad + \text{Tr} \left(s^{-1} [\tilde{\nabla}^{0,1} \Psi^{1,0}, s] s^{-1} [\Psi^{1,0}, s] - [s^{-1} \Psi^{1,0} s, s^{-1} \tilde{\nabla}^{0,1} s] s^{-1} [\Psi^{1,0}, s] \right) \\ &\quad + 2 \text{Tr} \left([\Psi^{0,1}, s^{-1} \Psi^{1,0} s] s^{-1} \tilde{\nabla}^{1,0} s \right) + 2 \text{Tr} \left(\Psi^{0,1} (s^{-1} [\Psi^{1,0}, s])^2 \right). \end{aligned} \quad (11)$$

We set

$$Z_{U, \lambda(k)}(\delta) = U \cap \{\tau_{\lambda(k)} = \delta\} \cap \bigcap_{\substack{1 \leq i \leq \ell \\ i \neq k}} \{\tau_{\lambda(i)} \geq \delta\}.$$

Let $[\eta] \in H^{n-2, n-2}(X, \mathbb{C})$. By using Lemma 4.13 and (11), we obtain

$$\begin{aligned} & \lim_{\delta \rightarrow 0} \int_{Z_{U, \lambda(k)}(\delta)} \left(2 \operatorname{Tr} (F_{\nabla_{H_1}} s^{-1} \nabla_{H_1}^{1,0} s) + \operatorname{Tr} (\bar{\partial} (s^{-1} \nabla_{H_1}^{1,0} s) \nabla_{H_1}^{1,0} s) \right) \cdot \eta \\ &= \lim_{\delta \rightarrow 0} \int_{Z_{U, \lambda(k)}(\delta)} \operatorname{Tr} \left(2 F_{\tilde{\nabla}}^{1,1} s^{-1} \tilde{\nabla}^{1,0} s + \tilde{\nabla}^{0,1} (s^{-1} \tilde{\nabla}^{1,0} s) s^{-1} \tilde{\nabla}^{1,0} s \right) \cdot \eta. \end{aligned} \quad (12)$$

We set $s_{\neq k} = \prod_{i \neq k} s_i$. It follows that $s = s_{\neq k} \cdot s_k = s_k \cdot s_{\neq k}$, and consequently:

$$s^{-1} \tilde{\nabla}^{1,0} s = s_{\neq k}^{-1} \tilde{\nabla}^{1,0} s_{\neq k} + s_k^{-1} \tilde{\nabla}^{1,0} s_k. \quad (13)$$

The limit expression becomes:

$$\begin{aligned} & \lim_{\delta \rightarrow 0} \int_{Z_{U, \lambda(k)}(\delta)} \operatorname{Tr} \left(\tilde{\nabla}^{0,1} (s^{-1} \tilde{\nabla}^{1,0} s) s^{-1} \tilde{\nabla}^{1,0} s \right) \eta \\ &= \lim_{\delta \rightarrow 0} \int_{Z_{U, \lambda(k)}(\delta)} \operatorname{Tr} \left(\left(\tilde{\nabla}^{0,1} (s_k^{-1} \tilde{\nabla}^{1,0} s_k) \right. \right. \\ & \quad \left. \left. + \tilde{\nabla}^{0,1} (s_{\neq k}^{-1} \tilde{\nabla}^{1,0} s_{\neq k}) \right) s_k^{-1} \tilde{\nabla}^{1,0} s_k \right) \cdot \eta. \end{aligned} \quad (14)$$

Define ${}^k\Gamma = \bigoplus (-a) \cdot \operatorname{id}_{G_a}$. The following identities hold:

$$s_k^{-1} \tilde{\nabla}^{1,0} (s_k) = {}^k\Gamma \cdot \partial \log \tau_k, \quad (15)$$

$$\tilde{\nabla}^{0,1} (s_k^{-1} \tilde{\nabla}^{1,0} (s_k)) = {}^k\Gamma (\bar{\partial} \partial \log \tau_k). \quad (16)$$

For ${}^k\operatorname{Gr}_a F_* = {}^kF_a / {}^kF_{>a}$, we have $\operatorname{Tr} {}^k\Gamma^2 = \sum_{a \in S} a^2 \operatorname{rank} ({}^k\operatorname{Gr}_a F_*)$. The limit evaluates to:

$$\begin{aligned} & \lim_{\delta \rightarrow 0} \int_{Z_{U, \lambda(k)}(\delta)} \operatorname{Tr} \left(\tilde{\nabla}^{0,1} (s_k^{-1} \tilde{\nabla}^{1,0} s_k) \cdot s_k^{-1} \tilde{\nabla}^{1,0} s_k \right) \cdot \eta \\ &= \pm \sum_a 2\pi \sqrt{-1} \int_{U \cap H_{\lambda(k)}} a^2 \operatorname{rank} ({}^k\operatorname{Gr}_a^F(E)) \bar{\partial} \partial (\log \tau_k) \cdot \eta. \end{aligned} \quad (17)$$

The C^∞ -decomposition $F|_{D_k} = \bigoplus {}^kG_a$ induces an isomorphism $F|_{D_k} \simeq \bigoplus {}^k\operatorname{Gr}_a F_*$. The restriction of $\tilde{\nabla}$ to $F|_{D_k \cap U}$ coincides with the Chern connection of $\bigoplus {}^k\operatorname{Gr}_a F_*$ under this isomorphism.

Let ${}^kH_{1,a}$ be the Hermitian metric on ${}^k\operatorname{Gr}_a F_*$ induced by H_1 . Then:

$$\begin{aligned} & \lim_{\delta \rightarrow 0} \int_{Z_{U, \lambda(k)}(\delta)} \operatorname{Tr} \left(F_{\tilde{\nabla}}^{1,1} s^{-1} \tilde{\nabla}^{1,0} s \right) \cdot \eta \\ &= \pm \sum_a 2\pi \sqrt{-1} \int_{U \cap H_{\lambda(k)}} (-a) \operatorname{Tr} F_{\nabla_{{}^kH_{1,a}}} \cdot \eta. \end{aligned} \quad (18)$$

Let ${}^k\tilde{H}_{1,a}$ be the Hermitian metric on ${}^k\operatorname{Gr}_a F_*|_{D_k^\circ}$ induced by $H_1 \cdot s_{\neq k}$. We derive:

$$\begin{aligned} & \lim_{\delta \rightarrow 0} \int_{Z_{U, \lambda(k)}(\delta)} \operatorname{Tr} \left(\left(F_{\tilde{\nabla}}^{1,1} + \tilde{\nabla}^{0,1} (s_{\neq k}^{-1} \tilde{\nabla}^{1,0} s_{\neq k}) \right) s^{-1} \tilde{\nabla}^{1,0} s \right) \cdot \eta \\ &= \pm \sum_a 2\pi \sqrt{-1} \int_{U \cap H_{\lambda(k)}} (-a) \operatorname{Tr} F_{{}^k\tilde{H}_{1,a}} \cdot \eta. \end{aligned} \quad (19)$$

Combining these results yields:

$$\begin{aligned}
\lim_{\delta \rightarrow 0} \pm \int_{Z_{U, \lambda(k)}(\delta)} & \left(2 \operatorname{Tr} (F_{\nabla_{H_1}} s^{-1} \nabla_{H_1}^{1,0} s) + \operatorname{Tr} (\bar{\partial} (s^{-1} \nabla_{H_1}^{1,0} s) s^{-1} \nabla_{H_1} s) \right) \cdot \eta \\
&= \sum_a 2\pi \sqrt{-1} \int_{U \cap H_{\lambda(k)}} (-a) \left(\operatorname{Tr} F_{\nabla_{kH_{1,a}}} + \operatorname{Tr} F_{\nabla_{k\tilde{H}_{1,a}}} \right) \cdot \eta \\
&\quad + \sum_a 2\pi \sqrt{-1} \int_{U \cap H_{\lambda(k)}} a^2 \operatorname{rank} ({}^k \operatorname{Gr}_a F_*) \bar{\partial} \partial (\log \tau_k) \cdot \eta. \quad (20)
\end{aligned}$$

Similarly, fix $1 \leq j, k \leq \ell$. Then we take the limit after integrating by part on the set

$$Z_{U, \lambda(j), \lambda(k)} := U \cap \{\tau_{\lambda(j)} = \tau_{\lambda(k)} = \delta\} \cap \bigcap_{\substack{1 \leq i \leq \ell \\ i \neq j, k}} \{\tau_{\lambda(i)} \geq \delta\}.$$

we could obtain

$$\begin{aligned}
\lim_{\delta \rightarrow 0} \pm \int_{Z_{U, \lambda(j), \lambda(k)}(\delta)} & \left(2 \operatorname{Tr} (F_{\nabla_{H_1}} s^{-1} \nabla_{H_1}^{1,0} s) + \operatorname{Tr} (\bar{\partial} (s^{-1} \nabla_{H_1}^{1,0} s) s^{-1} \nabla_{H_1} s) \right) \cdot \eta \\
&= \sum_{i \in [j, k]} \sum_a 2\pi \sqrt{-1} \int_{U \cap D_{\lambda(i)}} (-i a) \left(\operatorname{Tr} F_{\nabla_{iH_{1,a}}} + \operatorname{Tr} F_{\nabla_{i\tilde{H}_{1,a}}} \right) \cdot \eta \\
&\quad + \sum_{i \in [j, k]} \sum_a 2\pi \sqrt{-1} \int_{U \cap D_{\lambda(i)}} i a^2 \operatorname{rank} ({}^i \operatorname{Gr}_a F_*) \bar{\partial} \partial (\log \tau_{\lambda(i)}) \cdot \eta \\
&\quad - \sum_{a \in S^{(j, k)}} (2\pi)^2 \int_{U \cap D_{j, k}} {}^j a \cdot {}^k a \cdot \operatorname{rank} ({}^{jk} \operatorname{Gr}_a F_*) \cdot \eta \quad (21)
\end{aligned}$$

Combining equations (20) and (21) we get the adaptedness in codimension 2, i.e., the following proposition.

Proposition 4.14 $\operatorname{ch}_2(F_*) \cdot \eta = \frac{1}{2} \left(\frac{\sqrt{-1}}{2\pi} \right)^2 \int_{X \setminus D} \operatorname{Tr} (F_{\nabla_{H_0}} \wedge F_{\nabla_{H_0}}) \wedge \eta.$

Now return to our previous setting, i.e. a parabolic sheaf $\mathcal{F}_*$ over (X, D) . As we have remarked in the beginning of this subsection, we can construct a metric $\widehat{H}$ on the regular part of $\mathcal{F}_*$ which is adapted to it in codimension 2, i.e., the following proposition.

Proposition 4.15 $\widehat{H}$ is adapted to $\mathcal{F}_*$ in codimension 2.

Moreover, we can prove the following proposition by the blow-up technique used in Section 3. As it is tautological, we omit the proof.

Proposition 4.16 $\widehat{H}$ is adapted to any parabolic subsheaf S_* of $\mathcal{F}_*$ in codimension 1.

Following from Lemma 4.11. We also have the curvature bound for $\widehat{H}$.

Lemma 4.17 $|F_{\widehat{H}}|_{\widehat{H}} = O(|\sigma|^{-(1+\varepsilon)}).$

5 Parabolic stability and Analytic stability

Given a parabolic sheaf $\mathcal{F}_*$ over a compact Kähler manifold (X, ω) . We define the parabolic degree $\deg_{\omega}(\mathcal{F}_*)$ of $\mathcal{F}_*$ with respect to ω by

$$\deg_{\omega}(\mathcal{F}_*) := \operatorname{ch}_1(\mathcal{F}_*) \cdot \frac{[\omega]^{n-1}}{(n-1)!},$$

and the slope $\mu_\omega(\mathcal{F}_*)$ of $\mathcal{F}_*$ with respect to ω by

$$\mu_\omega(\mathcal{F}_*) := \frac{\deg_\omega(\mathcal{F}_*)}{\text{rank}(\mathcal{F}_*)}.$$

Definition 5.1 (Parabolic stability) If for any proper parabolic subsheaf $\mathcal{S}_*$ of $\mathcal{F}_*$, $\mu_\omega(\mathcal{S}_*) < \mu_\omega(\mathcal{F}_*)$, we say that $\mathcal{F}_*$ is parabolic stable with respect to ω .

Next, we recall Simpson's [32] definition of the analytic stability of a Hermitian holomorphic vector bundle $(E, \bar{\partial}, H)$ over a not necessarily compact Kähler manifold (X°, ω) . The analytic degree $\deg_{H, \omega}(\mathcal{S})$ of a torsion-free subsheaf $\mathcal{S}$ of E is given by

$$\deg_{H, \omega}(\mathcal{S}) := \frac{\sqrt{-1}}{2\pi} \int_{S_{reg}} \text{Tr}(F_{H|_{\mathcal{S}}}) \wedge \frac{\omega^{n-1}}{(n-1)!},$$

where $H|_{\mathcal{S}}$ is the restriction of H to the locally free part of $\mathcal{S}$ and S_{reg} is the Zariski open subset of X° where $\mathcal{S}$ is locally free and the analytic slope $\mu_{H, \omega}(\mathcal{S})$ is defined as

$$\mu_{H, \omega}(\mathcal{S}) := \frac{\deg_{H, \omega}(\mathcal{S})}{\text{rank}(\mathcal{S})}.$$

Notice that by Chern-Weil's formula we have

$$\deg_{H, \omega}(\mathcal{S}) = \frac{\sqrt{-1}}{2\pi} \int_{S_{reg}} \text{Tr}(\mathbf{p}_{\mathcal{S}, \hat{H}} \wedge_\omega F_H) - \frac{1}{2\pi} \int_{S_{reg}} |\bar{\partial} \mathbf{p}_{\mathcal{S}, \hat{H}}|_H^2$$

where the operator Λ_ω stands for the contraction with respect to ω and $\mathbf{p}_{\mathcal{S}}$ is the orthogonal projection of E onto $\mathcal{S}$. Hence if we want the above notions defined over the non-compact manifold X° to make sense, we need to require $F_H \in L^1(X^\circ, \omega)$ and $\mathbf{p}_{\mathcal{S}, \hat{H}} \in L_1^2(S_H)$ where $L_1^2(S_H)$ is the Sobolev space of sections of $\text{End}(E)$ that are self-adjoint with respect to H .

Definition 5.2 (Analytic stability) If for any proper saturated subsheaf $\mathcal{S}$ of E such that $\mathbf{p}_{\mathcal{S}, \hat{H}} \in L_1^2(S_H)$, $\mu_{H, \omega}(\mathcal{S}) < \mu_{H, \omega}(E)$, we say that E is analytic stable with respect to ω and H .

In the last section, we have constructed a metric $\hat{H}$ that is adapted to any parabolic subsheaf $\mathcal{S}_*$ of $\mathcal{F}_*$ in codimension 1. And $\hat{H}$ is defined on the vector bundle $F|_{X^\circ \setminus D}$ where F is the locally free part of the underlying sheaf $\mathcal{F}$. We should have the following proposition.

Proposition 5.1 *The parabolic stability of $\mathcal{F}_*$ with respect to ω is equivalent to the analytic stability of $F|_{X^\circ \setminus D}$ with respect to the restriction of ω and $\hat{H}$.*

Proof Suppose F is analytic stable. For any proper parabolic subsheaf $\mathcal{S}_*$ of $\mathcal{F}_*$, $\mathcal{S}|_{X^\circ \setminus D}$ is a torsion-free subsheaf of F . Then it follows immediately from Lemma 4.16 that $\mathcal{F}_*$ is parabolic stable.

Conversely, suppose $\mathcal{F}_*$ is parabolic stable, we need to compare the analytic slope of any proper saturated subsheaf $\mathcal{S}$ of $F|_{X^\circ \setminus D}$ such that $\mathbf{p}_{\mathcal{S}, \hat{H}} \in L_1^2(S_{\hat{H}})$ with the analytic slope of $F|_{X^\circ \setminus D}$. Again by Lemma 4.16, it suffices to show that any saturated subsheaf $\mathcal{S}$ of F with $\mathbf{p}_{\mathcal{S}, \hat{H}} \in L_1^2(S_{\hat{H}})$ can be extended to a parabolic subsheaf of $\mathcal{F}_*$. It was proved in [22, Proposition 5.9] that $\mathcal{S}$ can be extended to X° as a coherent subsheaf of F . Since $X - X^\circ$ is an analytic subset of codimension 3 and $\mathcal{F}$ is reflexive, it can be extended further to a coherent subsheaf $\mathcal{S}'$ of $\mathcal{F}$. Let $\mathcal{S}''$ be the saturation of $\mathcal{S}'$, then $\mathcal{S}''$ is a parabolic subsheaf of $\mathcal{F}_*$ extending $\mathcal{S}$. $\square$

For the convenience of the readers as well as the later use, we conclude the spirit of [22, Proposition 5.9] in the following proposition. As a preparation, we need the following concept concerning the singularities of a Hermitian metric on a divisor.

Definition 5.3 Let Δ^n be the polydisk centered at origin and Δ^* be the punctured unit disk. Let E be the trivial holomorphic vector bundle over Δ^n and H be a possibly singular metric well defined on $E|_{\Delta^{n-k} \times (\Delta^*)^k}$. We say that H is of polynomial growth along the divisor $z_{k+1} \cdot z_{k+2} \cdots z_n = 0$, if $|H| = O(|z'|^a)$ for some $a \in \mathbb{R}$, where $z' := (z_{k+1}, \dots, z_n)$. If $a > 0$, we also say that H is of polynomial decay.

Then [22, Proposition 5.9] says:

Proposition 5.2 *If H is an Hermitian metric defined on $F_{X^\circ \setminus D}$ that is locally of polynomial growth along D , then for any proper saturated coherent subsheaf $\mathcal{S}$ of $F_{X^\circ \setminus D}$ such that $p_{\mathcal{S}, \hat{H}} \in L_1^2(S_H)$, $\mathcal{S}$ can be extended as a coherent subsheaf of F over X° .*

6 Kobayashi-Hitchin Correspondence

We use the H-Y-M flow to deform $\hat{H}$. We will show that if the parabolic sheaf is stable (resp. semistable), then there is a H-E (resp. an approximate) structure on the bundle $F|_{X^\circ \setminus D}$ that is compatible with the parabolic structure. Since $\hat{H}$ is only defined on $X^\circ - D$ and we are not clear about the property of $\hat{H}$ near the singular locus W , we investigate the behavior of the H-Y-M flow on $\tilde{X} \setminus D$ first where $\hat{H}$ is a well-defined metric on $E|_{\tilde{X} \setminus D}$ with polynomial decay approaching D .

6.1 H-Y-M flow

For any conical Kähler metric $\omega_{\epsilon\delta}$ that we have constructed in Section 4 on $\tilde{X} \setminus D$, we consider the H-Y-M flow

$$H^{-1} \frac{dH}{dt} = -2 \left(\sqrt{-1} \Lambda_{\omega_{\epsilon\delta}} F_H - \lambda_{\epsilon\delta} \cdot \text{id}_{E'} \right) \quad (22)$$

living on $E|_{\tilde{X} \setminus D}$, where

$$\lambda_{\epsilon\delta} := \frac{\sqrt{-1}}{\text{Vol}(\tilde{X}, \omega_{\epsilon\delta})} \cdot \int_{\tilde{X} \setminus D^*} \text{Tr}(F_{\hat{H}}) \wedge \frac{\omega_{\epsilon\delta}^{n-1}}{(n-1)!}.$$

The following proposition was obtained by Simpson [32].

Proposition 6.1 *Let $(\tilde{X} \setminus D, \omega_{\epsilon\delta})$ satisfies the three assumptions mentioned in Proposition 4.3. Suppose $\hat{H}$ is a metric satisfies the assumption that $\|\Lambda_{\omega_{\epsilon\delta}} F_{\hat{H}}\|_{L^\infty(\hat{H})} \leq B$ where B is a positive constant. Then there is a unique solution $H(t)$ to the H-Y-M flow with $\det(H) = \det(\hat{H})$ such that $H(0) = \hat{H}$, such that $\|H\|_{L^\infty(\hat{H})}$ is bounded on each finite interval of time. For this solution, $\|\Lambda_{\omega_{\epsilon\delta}} F_H\|_{L^\infty(H)} \leq B$ for all t .*

It follows from Lemma 4.17 that, for sufficiently small δ , we have $\|\Lambda_{\omega_{\epsilon\delta}} F_{\hat{H}}\|_{L^\infty(\hat{H})} \leq B$. Hence the above proposition can be applied and we obtain a family of solutions $H_{\epsilon\delta}$ with the same initial value $\hat{H}$. We wish to show that $H_{\epsilon\delta}$ will converge to a solution of the H-Y-M flow

$$H^{-1} \frac{dH}{dt} = -2 \left(\sqrt{-1} \Lambda_\omega F_H - \lambda \cdot \text{id}_F \right) \quad (23)$$

living on the bundle $F := \mathcal{F}|_{X^\circ \setminus D}$ where

$$\lambda := \frac{2\pi \cdot \mu_\omega(\mathcal{F}_*)}{\text{Vol}(X, \omega)}.$$

Furthermore, under the stability assumption of the parabolic sheaf, we hope that the initial metric $\widehat{H}$ will deform into an H-E metric with respect to ω along the flow. The idea of the proof basically comes from [2]. Firstly, we do some estimates.

Lemma 6.2 $\|\Lambda_{\epsilon\delta} F_{\widehat{H}}\|_{L^1(\widehat{H}, \omega_{\epsilon\delta})} \leq C_6$ with C_6 independent of ϵ and δ , i.e. $\Lambda_{\epsilon\delta} F_{\widehat{H}}$ is uniformly integrable.

Proof It follows from Lemma 4.17 that we may fix an ϵ_1 and a δ_1 such that $\sqrt{-1} \text{Tr}(F_{\widehat{H}}) \leq C_7 \cdot \omega_{\epsilon_1\delta_1}$. Then it holds:

$$\begin{aligned} |\Lambda_{\epsilon\delta} F_{\widehat{H}}| \cdot \omega_{\epsilon\delta}^n &\leq \left(|\Lambda_{\epsilon\delta}(C_7 \cdot \omega_{\epsilon_1\delta_1} \cdot I - \sqrt{-1} F_{\widehat{H}})| + |C_7 \cdot \Lambda_{\epsilon\delta} \omega_{\epsilon_1\delta_1} \cdot I| \right) \cdot \omega_{\epsilon\delta}^n \\ &\leq n \cdot \text{Tr} \left(2 \cdot C_7 \cdot \omega_{\epsilon_1\delta_1} \cdot I - \sqrt{-1} F_{\widehat{H}} \right) \cdot \omega_{\epsilon\delta}^{n-1}. \end{aligned}$$

Integrating on both sides, we get

$$\begin{aligned} \int_{\widetilde{X} \setminus D} |\Lambda_{\epsilon\delta} F_{\widehat{H}}| \cdot \omega_{\epsilon\delta}^n &\leq n \cdot \int_{\widetilde{X} \setminus D} (2r C_7 \cdot \omega_{\epsilon_1\delta_1} - \sqrt{-1} \text{Tr}(F_{\widehat{H}})) \wedge \omega_{\epsilon\delta}^{n-1} \\ &\leq n \cdot \int_{\widetilde{X} \setminus D} (2r C_7 \cdot \omega_{1\delta_1} - \sqrt{-1} \text{Tr}(F_{\widehat{H}})) \wedge \omega_{1\delta}^{n-1} \\ &= n \cdot \int_{\widetilde{X} \setminus D} (2r C_7 \cdot \omega_1 - \sqrt{-1} \text{Tr}(F_{\widehat{H}})) \wedge \omega_1^{n-1} \\ &= C_6. \end{aligned}$$

□

Unless otherwise specified, the constants appear in the estimates in the rest of the paper will always be uniform in ϵ and δ .

Along the heat flow (22), we have the following inequalities (c.f. [10]):

$$\begin{aligned} \left(\Delta_{\epsilon\delta} + \frac{\partial}{\partial t} \right) |\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}}|_{H_{\epsilon\delta}} &\leq 0 \\ \left(\Delta_{\epsilon\delta} + \frac{\partial}{\partial t} \right) |\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}}|_{H_{\epsilon\delta}}^2 &\leq 0. \end{aligned}$$

And we put $f(t) := |\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}}|_{H_{\epsilon\delta}}$.

Lemma 6.3 $\|f_t\|_{L^1}$ and $\|f_t\|_{L^2}$ is non-increasing with time.

Proof Suppose to the contrary that there is $t_2 > t_1$ such that $\|f(t_2)\|_{L^1} = \|f(t_1)\|_{L^1} + \delta$ with $\delta > 0$. Since $\|f_t\|_{L^\infty} \leq B$ and $|dV_{\omega_{\epsilon\delta}}| < \infty$, we can take a relatively compact region $Z \subset X$ such that $B|dV_{\omega_{\epsilon\delta}}|(X - Z) < \frac{\delta}{8}$. Then we have

$$\|f(t_2)\|_{L^1(Z)} \geq \|f(t_1)\|_{L^1(Z)} + \frac{3}{4}\delta.$$

On the other hand, let Ω^φ be a family of nested compact regions with smooth boundaries whose limit exhausts $X^k - D^*$. Simpson [32, Section 6] showed that the solution $H_{\epsilon\delta}$ can be obtained by taking the C_{loc}^∞ limit of a sequence of metrics $H_{\epsilon\delta}^\varphi$ which are the solutions to

the H-Y-M flow (22) over Ω^φ satisfying the Neumann boundary condition. We put $f^\varphi(t) := |\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}^\varphi}|_{H_{\epsilon\delta}^\varphi}$. Then we have

$$\left(\Delta_{\epsilon\delta} + \frac{\partial}{\partial t}\right) f^\varphi \leq 0.$$

Integrating by part on both side over Ω^φ and using the Neumann boundary condition, we obtain that $\frac{\partial}{\partial t} \|f^\varphi\|_{L^1(\Omega^\varphi)} \leq 0$ for any $t > 0$. As $H_\varphi(t_i)$ converges in $C^\infty(Z)$ to $H(t_i)$ where $i = 1, 2$ as $\varphi \rightarrow \infty$, hence for φ sufficiently large, we have

$$|||f^\varphi(t_i)|||_{L^1(Z)} - \|f(t_i)\|_{L^1(Z)} \leq \frac{\delta}{8}.$$

But

$$\|f(t_2)^\varphi\|_{L^1(X_\varphi)} \leq \|f(t_1)^\varphi\|_{L^1(X_\varphi)},$$

hence

$$\|f(t_2)^\varphi\|_{L^1(Z)} \leq \|f(t_1)^\varphi\|_{L^1(Z)} + \frac{\delta}{4},$$

therefore

$$\|f(t_2)\|_{L^1(Z)} \leq \|f(t_1)\|_{L^1(Z)} + \frac{1}{2}\delta,$$

thus a contradiction. The same reasoning applies to $\|f_t\|_{L^2}$. $\square$

As a corollary, we have:

Lemma 6.4

$$\|\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}}\|_{L^1(H_{\epsilon\delta})} \leq C_6$$

for $t \in [0, \infty[$.

Lemma 6.5 For any $t > 0$, $\|\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}}\|_{L^\infty(H_{\epsilon\delta}, \omega_{\epsilon\delta})} \leq C_8(\max(t^{-1}, 1))$.

Proof Recall that we have derived a Sobolev inequality for smooth functions compactly supported on the measure space $(\tilde{X} \setminus D^*, \omega_{\epsilon\delta})$ in Lemma 4.5 and the Sobolev constant is independent of ϵ and δ . We may consider the function

$$f_\varsigma := |\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}}|_{H_{\epsilon\delta}} + \varsigma(\log(|\sigma|^2) - At)$$

where A is set as a large number to make sure that $f_{\varsigma\nu}$ is a subsolution to the heat equation. Then $\varphi := \eta_a(t)^2 \cdot (f_{\varsigma\nu}^+)^{2a-1}$ with $a \geq 1$ is a legitimate test function for the parabolic Moser's iteration technique. Here $\eta_a(t)$ is an appropriate cutoff function. We obtain for any $T > 0$,

$$\|f_\varsigma(t)\|_{L^\infty((\tilde{X} \setminus D) \times [1.5T, 2T])} \leq C_8(T^{-1}) \|f_\varsigma(t)\|_{L^1((\tilde{X} \setminus D) \times [T, 2T])}.$$

Then the lemma follows by taking the limits on both sides as $\varsigma \rightarrow 0$. $\square$

We put $B_{\omega_1}(d) := \{x \in \tilde{X} \mid d_{\omega_1}(x, \mathcal{E}) < d\}$. Then it follows easily from Lemma 4.17 that

$$|\Lambda_{\omega_{\epsilon\delta}} F_{\hat{H}}|_{\hat{H}} \leq C_9(d^{-1})|\sigma|^{-2}$$

over $B_{\omega_1}(d)^c$ with $C_9(d^{-1})$ independent of ϵ and δ . And we have the following lemma.

Lemma 6.6 For any $t \geq 0$, there exists a constant $C_{10}(d^{-1})$ such that:

$$|\Lambda_{\omega_{\epsilon\delta}} F_{H_{\epsilon\delta}}|_{H_{\epsilon\delta}} \leq C_{10}(d^{-1})|\sigma|^{-2}$$

for all $(x, t) \in B_{\omega_1}(d)^c \times [0, \infty[$.

Proof The crucial part of the proof is the uniform Gaussian upper bound of the heat kernel $K_{\epsilon\delta}$. But it is obtained in Proposition 4.6. The rest of the proof follows in the same way as in the proof in [24, Lemma 2.2]. $\square$

We put $h_{\epsilon\delta} := H_{\epsilon\delta} \widehat{H}^{-1}$. We know that $|H_{\epsilon\delta}|_{\widehat{H}}$ and $|H_{\epsilon\delta}^{-1}|_{\widehat{H}}$ is comparable to the positive quantity $\Phi_{\epsilon\delta} = \log(h_{\epsilon\delta}) + \log(h_{\epsilon\delta}^{-1}) - 2 \operatorname{rank}(E)$. And we have

$$\frac{\partial}{\partial t} \Phi_{\epsilon\delta} \leq 2(|\Lambda_{\omega_{\epsilon\delta}} F_{H_{\epsilon\delta}}|_{H_{\epsilon\delta}} + |\Lambda_{\omega_{\epsilon\delta}} F_{\widehat{H}}|_{\widehat{H}}) \quad (24)$$

Hence we obtain:

Lemma 6.7 For $(x, t) \in B_{\omega_1}(d)^c \times [0, T]$,

$$\begin{aligned} |H_{\epsilon\delta}|_{\widehat{H}} &\leq TC_{11}(d^{-1})|\sigma|^{-2} \\ |H_{\epsilon\delta}^{-1}|_{\widehat{H}} &\leq TC_{11}(d^{-1})|\sigma|^{-2} \end{aligned}$$

To get the convergence of the H-Y-M flow, we need one more proposition of [2, Proposition 1].

Proposition 6.8 Let H be an Hermitian matrix valued function defined on a Kähler manifold (Y, ω) which belongs to the Sobolev space L^2_1 . Assume that H and H^{-1} is uniformly bounded and it satisfies the elliptic equation

$$\Lambda_{\omega} \bar{\partial}(\partial H H^{-1}) = f$$

in a weak sense with a uniformly bounded function f . Then H belongs to $C^{1,\alpha}_{loc}$ for any $0 < \alpha < 1$ and admits an estimate depending only on $\|H\|_{L^\infty}$, $\|H^{-1}\|_{L^\infty}$, $\|f\|_{L^\infty}$ and the geometry of Y .

Now we can apply the above proposition to do the interior estimates for $H_{\epsilon\delta}$ over $(\widetilde{X}^\circ - D, \omega_{\epsilon\delta})$. Notice that $\omega_{\epsilon\delta}$ is locally uniformly quasi-isometric to the fixed Kähler metric ω_1 . Then we simply set $\Lambda_{\omega_{\epsilon\delta}} F_{H_{\epsilon\delta}}$ as the f in the above proposition. Hence $H_{\epsilon\delta}$ is uniformly bounded in $C^{1,\alpha}_{loc}$ -topology for any $t \geq 0$. On the other hand, by the H-Y-M flow, we can see that $\frac{dH_{\epsilon\delta}}{dt}$ is uniformly bounded in C^0_{loc} -topology for any $t > 0$. Hence $H_{\epsilon\delta}$ converges in $C^{1/0}_{loc}$ -topology to a time flow of Hermitian metric H living on $E|_{\widetilde{X} \setminus D^* - \pi^{-1}(W)}$, or equivalently on $F := \mathcal{F}_{X \setminus (D \cup W)}$ with the initial value $\widehat{H}$. And we may apply the parabolic Schauder estimate to show that H is indeed a smooth solution and $H_{\epsilon\delta}$ converges in $C^{\infty/\infty}_{loc}$ -topology to H . Moreover, due to Lemma (24), for any $t > 0$, we have that $H(t)$ is locally of polynomial decay approaching D .

6.2 Correspondence

From stability to H-E metric Based on the H-Y-M flow $H(t)$, we want to show that under the stability condition of the parabolic sheaf $\mathcal{F}_*$, H will converge to an H-E metric on $X^\circ \setminus D$ which is compatible with the parabolic structure. Parallely, we also want to show that under the semistability condition, $H(t)$ will give us a family of approximate H-E metrics all of which are compatible with the parabolic structure.

Indeed, the difficulty comes from the fact that $|\Lambda F_{\widehat{H}}|_{\widehat{H}}$ is not bounded on $X^\circ \setminus D$. Hence Simpson's [32] arguments cannot be directly applied. One may argue that one can consider the H-Y-M flow starting from a positive time point and then apply Simpson's results. Although from the above analyses, we know that $|\Lambda F_{H(t)}|_{H(t)}$ is bounded for any $t > 0$, but it seems

difficult to show the analytic stability of $F|_{X^\circ \setminus D}$ with respect to $H(t)$. But this dilemma was resolved by the second author in [24, Proposition 4.1], which shows that under the semistability condition, we still have

$$\lim_{t \rightarrow \infty} \|\Phi(t)\|_{L^2(H(t))} = 0$$

where $\Phi(t) := \|\sqrt{-1} \cdot \Lambda F_{H(t)} - \lambda \cdot \text{id}_F\|_{L^2(H(t))}$. This implies the existence of approximate H-E metrics. Indeed, we have

$$\left(\frac{\partial}{\partial t} + \Delta_\omega\right) \cdot \Phi(t)^2 \leq 0.$$

Then we choose a test function for Moser's iteration as in Lemma 6.5 to get

$$\|\Phi(t)\|_{L^\infty(H(t))(X \setminus D \times [1.5T, 2T])} \leq C_{12} \|\Phi(t)\|_{L^2(H(t))(X \setminus D \times [T, 2T])}.$$

For the stable case, the same trick used in [24, Proposition 4.1] can be applied to show that Proposition 5.3 of [32] holds for $H(t)$ which is the crucial estimate calling for the stability condition. Then the arguments in Section 7 of [32] can be applied to show that $H(t)$ converges to an H-E metric $H(\infty)$.

Hence we only need to show that the metrics (H-E or approximate H-E) are compatible with the parabolic structure and admissible in the sense of Definition 1.1. It suffices to prove for the approximate H-E metrics as the H-E case follows directly from Fatou's lemma.

We need a lemma first.

Lemma 6.9 (Lemma 5.2 in [32]) *Suppose Y is a noncompact Kähler manifold which has an exhaustion function ϕ with $\int_Y |\Delta \phi| < \infty$, and suppose η is a $(2n-1)$ -form with $\int_Y |\eta|^2 < \infty$. Then if $d\eta$ is integrable, $\int_Y d\eta = 0$.*

Let us fix a Kähler metric ω_c on $(\tilde{X} \setminus D)$ with cusp singularities along D . It follows that the density functions $\frac{\omega_{\epsilon\delta}^n}{\omega_c^n}$ are uniformly bounded in ϵ and δ .

Proposition 6.10 *If $\mathcal{F}_*$ is semistable, then for any $t > 0$, $H(t)$ is compatible with the parabolic structure.*

Proof As $\det(\hat{H}) = \det(H(t))$, $H(t)$ is adapted to $\mathcal{F}_*$ in codimension 1. Hence it suffices to show that

$$\sqrt{-1} \cdot \int_{X^\circ \setminus D} \text{Tr}(F_{\hat{H}|_S}) \wedge \omega^{n-1} \leq \sqrt{-1} \cdot \int_{X^\circ \setminus D} \text{Tr}(F_{H(t)|_S}) \wedge \omega^{n-1}$$

for any proper parabolic subsheaf $\mathcal{S}_*$.

By the blow-up procedure in Section 3, it can be shown that

$$\int_{\tilde{X}^\circ \setminus D^*} \text{Tr}(F_{\hat{H}|_S}) \wedge \omega^{n-1} = \int_{X^\circ \setminus D} \text{Tr}(F_{\hat{H}|_S}) \wedge \omega^{n-1}.$$

An adaptation of Lemma 6.2 yields the following bound:

$$\left| \Lambda_{\epsilon\delta} F_{\hat{H}|_S} \right| \cdot \omega_{\epsilon\delta}^n \leq n \cdot \text{Tr} \left(2 \cdot C_7 \cdot \omega_{1\delta_1} \cdot I - \sqrt{-1} F_{\hat{H}|_S} \right) \cdot \omega_{1\delta}^{n-1}.$$

Since the integral of the right-hand side is finite and independent of δ , then by generalized dominated convergence theorem, we obtain

$$\int_{X^\circ \setminus D} \text{Tr}(F_{\hat{H}|_S}) \wedge \omega^{n-1} = \lim_{\substack{\epsilon \rightarrow 0 \\ \delta \rightarrow 0}} \int_{\tilde{X}^\circ \setminus D^*} \text{Tr}(F_{\hat{H}|_S}) \wedge \omega_{\epsilon\delta}^{n-1}.$$

For fixed ϵ, δ and t , we put $h_{\epsilon\delta|S} := H_{\epsilon\delta}(t)|_S \cdot \widehat{H}_{|S}^{-1}$ and $h_{\epsilon\delta} := H_{\epsilon\delta}(t) \cdot \widehat{H}^{-1}$. Simpson [32] showed that $\widehat{H}$ and $H_{\epsilon\delta}(t)$ are mutually bounded and $\bar{\partial}h_{\epsilon\delta} \in L^2(\widehat{H}, \omega_{\epsilon\delta})$. Hence $h_{\epsilon\delta}$ is bounded with respect to either $\widehat{H}$ or $H_{\epsilon\delta}(t)$.

Since we have

$$h_{\epsilon\delta|S} = p_{S, \widehat{H}} \cdot h_{\epsilon\delta} \cdot \iota \quad (25)$$

$$p_{S, H_{\epsilon\delta}(t)} = h_{\epsilon\delta|S}^{-1} \cdot p_{S, \widehat{H}} \cdot h_{\epsilon\delta} \quad (26)$$

where ι is the injection of S into E . Hence,

$$\bar{\partial}h_{\epsilon\delta|S} = \bar{\partial}p_{S, \widehat{H}} \cdot h_{\epsilon\delta} \cdot \iota + p_{S, \widehat{H}} \cdot \bar{\partial}h_{\epsilon\delta} \cdot \iota$$

which implies that $\bar{\partial}h_{\epsilon\delta|S} \in L^2(\widehat{H}, \omega_{\epsilon\delta})$. Then by Chern-Weil's formula and (26),

$$\text{Tr}(F_{H_{\epsilon\delta|S}}) \wedge \omega_{\epsilon\delta}^{n-1}$$

is integrable.

On the other hand,

$$\text{Tr}(F_{H_{\epsilon\delta|S}}) - \text{Tr}(F_{\widehat{H}_{|S}}) = \text{Tr}(\bar{\partial}(\partial\widehat{H}_{|S} h_{\epsilon\delta|S} \cdot h_{\epsilon\delta|S}^{-1})).$$

But the above reasoning shows that

$$\text{Tr}(\partial\widehat{H}_{|S} h_{\epsilon\delta|S} \cdot h_{\epsilon\delta|S}^{-1}) \wedge \omega_{\epsilon\delta}^{n-1} \in L^2(\widehat{H}, \omega_{\epsilon\delta}).$$

Then it follows from Lemma 6.9 that

$$\int_{\widetilde{X}^\circ \setminus D^*} \text{Tr}(F_{\widehat{H}_{|S}}) \wedge \omega_{\epsilon\delta}^{n-1} = \int_{\widetilde{X}^\circ \setminus D^*} \text{Tr}(F_{H_{\epsilon\delta|S}}) \wedge \omega_{\epsilon\delta}^{n-1}.$$

By Chern-Weil's formula again, we have

$$\sqrt{-1} \cdot \text{Tr}(F_{H_{\epsilon\delta|S}}) \wedge \omega_{\epsilon\delta}^{n-1} \leq \sqrt{-1} \text{Tr}(p_{S, H_{\epsilon\delta}} \Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}}) \cdot \frac{\omega_{\epsilon\delta}^n}{\omega_c^n} \cdot \omega_c^n.$$

But the term on the right-hand side is uniformly bounded as both of $|\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}}|_{H_{\epsilon\delta}}$ and $\frac{\omega_{\epsilon\delta}^n}{\omega_c^n}$ are uniformly bounded. Hence if we take the limit on both sides and apply Fatou's lemma, we obtain the desired inequality:

$$\sqrt{-1} \cdot \int_{X^\circ \setminus D} \text{Tr}(F_{\widehat{H}_{|S}}) \wedge \omega^{n-1} \leq \sqrt{-1} \cdot \int_{X^\circ \setminus D} \text{Tr}(F_{H(t)|S}) \wedge \omega^{n-1}.$$

□

Proposition 6.11 *For any $t > 0$, $H(t)$ is admissible.*

Proof By Lemma 6.5, $|\Lambda F_H|_H$ is bounded on $X \setminus D$ for any $t > 0$. And we have the following:

$$\begin{aligned}
 & -8\pi^2 \int_X \text{ch}_2(\mathcal{F}_*) \wedge \frac{\omega^{n-2}}{(n-2)!} \\
 &= \lim_{\substack{\epsilon \rightarrow 0 \\ \delta \rightarrow 0}} \int_{\tilde{X}} \text{tr} (F_{\hat{H}} \wedge F_{\hat{H}}) \wedge \frac{\omega_{\epsilon\delta}^{n-2}}{(n-2)!} \\
 &\geq \lim_{\substack{\epsilon \rightarrow 0 \\ \delta \rightarrow 0}} \int_{\tilde{X}} \text{tr} (F_{H_{\epsilon\delta}(t)} \wedge F_{H_{\epsilon\delta}(t)}) \wedge \frac{\omega_{\epsilon\delta}^{n-2}}{(n-2)!} \quad (27) \\
 &= \lim_{\substack{\epsilon \rightarrow 0 \\ \delta \rightarrow 0}} \int_{\tilde{X}} \left(|F_{H_{\epsilon\delta}(t)}|_{H_{\epsilon\delta}(t), \omega_{\epsilon\delta}}^2 - |\Lambda_{\epsilon\delta} F_{H_{\epsilon\delta}(t)}|_{H_{\epsilon\delta}(t)}^2 \right) \cdot \frac{\omega_{\epsilon\delta}^n}{\omega_c^n} \cdot \frac{\omega_c^n}{n!} \\
 &\geq \int_X \left(|F_{H(t)}|_{H(t), \omega}^2 - |\sqrt{-1} \Lambda_\omega F_{H(t)}|_{H(t)}^2 \right) \cdot \frac{\omega^n}{n!}
 \end{aligned}$$

where the first inequality follows from [22, Lemma 7.1] and the second inequality follows by applying Fatou's lemma with respect to the measure induced by ω_c . Hence for any $t > 0$, $F_{H(t)}$ is square integrable. $\square$

From H-E metric to stability So far, we have done one direction of the Kobayashi-Hitchin correspondence.

To prove the converse part, suppose $H(t)$ is a family of approximate H-E metrics compatible with the parabolic structure, then by definition and Chern-Weil's formula, for any proper subsheaf $\mathcal{S}_*$, we have

$$\begin{aligned}
 \mu_\omega(\mathcal{S}_*) &\leq \liminf_{t \rightarrow \infty} \frac{\sqrt{-1}}{2\pi \text{rank}(\mathcal{S})} \int_{X \setminus D} \text{Tr}(F_{H|_{\mathcal{S}(t)}}) \wedge \omega^{n-1} \\
 &\leq \lim_{t \rightarrow \infty} \frac{\sqrt{-1}}{2\pi \text{rank}(\mathcal{F})} \int_{X \setminus D} \text{Tr}(F_{H(t)}) \wedge \omega^{n-1} \\
 &= \mu_\omega(\mathcal{F}_*).
 \end{aligned}$$

hence we are done with the semistable case.

To prove the converse part for the polystable case, we need a proposition first, which is a slight generalization of [2, Theorem 2 b)].

Proposition 6.12 *Let $(\Delta^n, z_1, \dots, z_n)$ be a polydisk and D be a divisor defined by $z_1 \cdot z_2 \cdots z_m = 0$. Let S be a closed subset with locally finite Hausdorff measure of real codimension 4. Let $\mathcal{F}_*$ be a saturated reflexive parabolic sheaf on Δ^n whose regular part F is defined on $\Delta^n \setminus (D \cap S)$. Suppose $\mathcal{F}_*$ admits an admissible H-E metric H on F which is compatible with the parabolic structure. Then for any local section s of $\mathcal{F}$, $H(s, s)$ is locally bounded and it belongs to $L_{1, \text{loc}}^2$.*

Proof The proof basically follows from [2, Section 1].

For a projection p from Δ^n to Δ^{n-2} along a generic direction, the set $S \cap p^{-1}(0)$ consists of a countable number of points which may accumulate only at 0. And there is a compact subset K of Δ^2 such that S is contained in $K \times \Delta^{n-2}$.

We put $X_t := p^{-1}(t)$. Then except for t in a subset of measure zero of Δ^{n-2} , we have

- $S_t := X_t \cap S$ contains only finite points.
- $D_t := D \cap p^{-1}(t)$ is a simple normal crossing divisor.

We put $u := \log^+(H(s, s))$ and $u_t := u|_{X_t}$. We analyse u_t within each slice X_t .

Suppose x_0 is an isolated point of S_t away from D_t . Then it is proved in [1, Section 3] that u_t belongs to L^2_1 near x_0 and satisfies the following inequality weakly

$$\Delta_t u_t \leq 4|F_t|.$$

If x_0 is contained in D_t , then without loss of generality, we may choose a local coordinate neighborhood (U_{x_0}, z_1, z_2) in X_t centered at x_0 such that D_t is defined by $z_1 \cdot z_2 = 0$. Due to the fact that $\mathcal{F}_*$ is a saturated reflexive parabolic sheaf and the slice U_{x_0} is of complex dimension 2, we know that $\mathcal{F}_{*|U_{x_0}}$ is a parabolic bundle.

On the other hand, $F|_{U_{x_0} \setminus D_t}$ admits an admissible H-E metric H_t . A regularity theorem in real dimension 4 with singularities in real codimension 2 (cf. [31]) from gauge theory implies that if the curvature tensor F_{H_t} of a H-E metric belongs to L^2 , then it belongs to L^p for some $p > 2$. Then O.Biquard [4, Theorem 2.1] proved that $F|_{U_{x_0} \setminus D_t}$ can be uniquely extended as a parabolic bundle F_* to U_{x_0} by choosing an appropriate complex gauge. In particular, for any holomorphic section s of F_* , $|s|_{H_t}$ is bounded. By the uniqueness of the extension, we know that $F_* \cong \mathcal{F}_{*|U_{x_0}}$. Hence, in this case, we also have the weak inequality

$$\Delta_t u_t \leq 4|F_t|.$$

Remark 6.13 Biquard was dealing with the extension problem in the case of a smooth divisor. The choice of a complex gauge involves solving a $\bar{\partial}$ -problem on a punctured polydisk $\Delta^* \times \Delta$. But the proof could be easily adapted to the case of a simple normal crossing divisor, i.e., solving the same $\bar{\partial}$ -problem on $\Delta^* \times \Delta^*$.

The remainder of the proof proceeds verbatim as in [2, Section 1]. Indeed, for any compact region $K' \subset \Delta^{n-2}$ containing 0, we can use the above inequality to show that $\nabla_t u$ is square integrable over $K \times K'$ where ∇_t is the gradient in the direction of the projection. As the direction of projection is generic, we have $u \in L^2_{1,loc}(\Delta^n)$. Once this is known, it is easy to see that $u \in L^\infty_{loc}$ as $\Delta F_H \in L^\infty_{loc}$ in view of

$$\Delta u \leq 2|\Delta F_H|.$$

□

Now suppose $\mathcal{F}_*$ is a saturated reflexive parabolic sheaf that admits an admissible H-E metric H compatible with the parabolic structure. Then it follows easily from the definition of H and Chern-Weil's formula that for any proper parabolic subsheaf $\mathcal{S}_*$, we have $\mu_\omega(\mathcal{S}_*) \leq \mu_\omega(\mathcal{F}_*)$.

Suppose the equality holds. We put $\mathcal{G}_* := \wedge^k \mathcal{F}_* \otimes \det(\mathcal{S}_*)^{-1}$. As $\det(\mathcal{S}_*)^{-1}$ is a parabolic line bundle, there exists an admissible H-E metric H' compatible with its parabolic structure. Hence the regular part of $\mathcal{G}$ inherits an admissible H-E metric H'' from H and H'

$$\lambda(\mathcal{G}) = \frac{2k\pi}{\text{Vol}(X, \omega)} (\mu_\omega(\mathcal{F}_*) - \mu_\omega(\mathcal{S}_*)) = 0.$$

On the other hand, by the above proposition, we know that $|s|''_H$ is bounded on X and it belongs to $L^2_1(X)$. Therefore, we have

$$\Delta_\omega |s|_H''^2 \leq 2\lambda(\mathcal{G}) |s|_H''^2 - 2|\bar{\partial}s|_H''^2 \leq 0$$

which implies $s \equiv C \neq 0$. Hence we have the holomorphic splitting:

$$\mathcal{F}|_{X^\circ} = \mathcal{S}|_{X^\circ} \oplus \mathcal{Q}|_{X^\circ}$$

where $\mathcal{Q}$ is subsheaf of $\mathcal{F}$.

Since W has codimension at least 3, we have

$$\mathrm{Ext}_1(\mathcal{S}, \mathcal{Q}) \cong \mathrm{Ext}_1(\mathcal{S}|_{X^\circ}, \mathcal{Q}|_{X^\circ}).$$

Hence we have

$$\mathcal{F}_* = \mathcal{S}_* \oplus \mathcal{Q}_*.$$

As a consequence, we have established the Kobayashi-Hitchin correspondence:

Theorem 6.14 *A saturated reflexive parabolic sheaf $\mathcal{F}_*$ over (X, ω, D) is μ_ω -polystable if and only if there exists an admissible H-E metric with respect to ω on $F_*|_{X^\circ \setminus D}$ which is compatible with $\mathcal{F}_*$.*

Theorem 6.15 *A saturated reflexive parabolic sheaf $\mathcal{F}_*$ over (X, ω, D) is μ_ω -semistable if and only if there exists a family of approximate H-E metrics with respect to ω on $F_*|_{X^\circ \setminus D}$ all of which are compatible with $\mathcal{F}_*$.*

Corollary 6.16 *If $\mathcal{F}_*$ is μ_ω -semistable with respect to ω , then*

$$\Delta(\mathcal{F}_*) \cdot [\omega]^{n-2} \geq 0.$$

Moreover, if $\mathcal{F}_$ is polystable, then the equality holds if and only if $\mathcal{F}|_{X \setminus D}$ is a vector bundle with a projectively flat H-E connection compatible with the parabolic structure of $\mathcal{F}_*$.*

Proof An obvious modification to the calculations in (27) yields the proof. $\square$

7 Bogomolov-Gieseker inequality for nef and big class

In this section, we prove a more general Bogomolov-Gieseker type inequality, i.e. the following theorem.

Theorem 7.1 *Let $\mathcal{F}_*$ be a saturated reflexive parabolic sheaf over a compact Kähler manifold (X, ω) which is semistable with respect to a nef and big class $[\eta]$. Then the Bogomolov-Gieseker inequality with respect to η holds, i.e.*

$$\Delta(\mathcal{F}_*) \cdot [\eta]^{n-2} \geq 0.$$

Let us briefly recall the definitions of nefness and bigness.

Definition 7.1 A class $[\eta] \in H^{k,k}(X, \mathbb{R})$ is called nef if for any $\epsilon > 0$, there exists a representative $\eta_\epsilon \in [\eta]$ such that $\eta_\epsilon \geq -\epsilon\omega^k$. It is clear that these classes form a closed cone in $H^{k,k}(X, \mathbb{R})$ which we denote as $\mathcal{N}^k$.

Definition 7.2 A class $[\eta] \in H^{k,k}(X, \mathbb{R})$ is called big if there exists a constant ϵ such that $\eta \geq \epsilon[\omega^k]$ in the sense of current. It is clear that these classes form an open cone in $H^{k,k}(X, \mathbb{R})$ which we denote as $\mathcal{B}^k$.

We put $[\eta_\epsilon] := [\eta + \epsilon\omega]$ where ω is a Kähler class. We have the following lemma concerning the openness of stability of $\mathcal{F}_*$ with respect to $[\eta_\epsilon]$.

Lemma 7.2 Suppose a parabolic sheaf $\mathcal{F}_*$ (not necessarily reflexive) is stable with respect to a class $[\eta] \in \mathcal{B}^1$. Then $\mathcal{F}_*$ is stable with respect to $[\eta_\epsilon]$ for any sufficiently small ϵ .

Proof Let $\mathcal{S}_*$ be a proper parabolic subsheaf of $\mathcal{F}_*$. We have

$$\mu_{\eta_\epsilon}(\mathcal{F}_*) - \mu_{\eta_\epsilon}(\mathcal{S}_*) = \mu_\eta(\mathcal{F}_*) - \mu_\eta(\mathcal{S}_*) + \epsilon \cdot \phi(\epsilon).$$

If we can show that

1. $|\phi(\epsilon)| \leq C_{12}$,
2. $\mu_\eta(\mathcal{F}_*) - \mu_\eta(\mathcal{S}_*) \geq C_{13}$

with C_{12}, C_{13} independent of the choices of $\mathcal{S}_*$ and ϵ , we are done.

We show the uniform bound of $\phi(\epsilon)$ first. As $\text{ch}_1(\mathcal{S}_*)$ is a linear combination of $\text{ch}_1(\mathcal{S})$ and D_i 's, hence $\phi(\epsilon)$ is a linear combination of the intersection numbers of $\text{ch}_1(\mathcal{F})$, $\text{ch}_1(\mathcal{S})$ and D_i 's with some bounded big classes parametrized by ϵ . Thus it suffices to show the following assertion:

Given a family of bounded big classes γ_ϵ , $\text{ch}_1(\mathcal{S}) \cdot [\gamma_\epsilon]$ has an uniform upper bound if we vary $\mathcal{S}$ and ϵ .

Following the spirit of Section 4, we can construct a metric H on the regular part of a torsion-free sheaf $\mathcal{F}$ which is adapted to any torsion free subsheaf $\mathcal{S}$ in codimension 1 (although we were dealing with a parabolic sheaf there, the idea of course could be transplanted to the ordinary sheaf case). Then by Gauss-Codazzi formula, we have

$$\text{ch}_1(\mathcal{S}) = \frac{\sqrt{-1}}{2\pi} \text{Tr}(\text{p}_{\mathcal{S}, H_0} \cdot F_{H_0}) - \text{Tr}((\bar{\partial} \text{p}_{\mathcal{S}, H_0})^\dagger \wedge \bar{\partial} \text{p}_{\mathcal{S}, H_0})$$

in the sense of current. Then it is not difficult to see that $\text{ch}_1(\mathcal{S}) \cdot [\gamma_\epsilon]$ is uniformly bounded.

The proof will be completed by the following lemma which implies 2. $\square$

Lemma 7.3 Let $\mathcal{F}_*$ be a parabolic sheaf and $[\eta] \in \mathcal{B}^1$. We put $\mu_\eta^* := \sup\{\mu_\eta(\mathcal{S}_*) | \mathcal{S}_* \subsetneq \mathcal{F}_*\}$. Then μ_η^* can be achieved by some proper subsheaf $\mathcal{S}_*^*$.

Proof It follows from [7, Lemma 2.2] that we can express $\eta^{n-1} = \sum_{i=1}^s \lambda_i \cdot e_i$ with $\lambda_i \geq 0$, $e_i \in H^{2(n-1)}(X, \mathbb{Q})$ and each e_i can be represented by a strictly positive current (not necessarily an $(n-1, n-1)$ -current). Then we have

$$\text{ch}_1(\mathcal{S}_*) \cdot [\eta^{n-1}] = \sum_{i=1}^s \lambda_i \cdot \text{ch}_1(\mathcal{S}_*) \cdot [e_i].$$

Since we want to achieve the maximal slope, we may restrict ourselves to the set of subsheaves with $-C_{14} < \deg_\eta(\mathcal{S}_*)$ with C_{14} a positive constant. By the arguments in the above lemma, we may choose the C_{14} large enough such that $\text{ch}_1(\mathcal{S}_*) \cdot [\eta^{n-1}] < C_{14}$ for all subsheaves. We put $\mathcal{A} := \{\mathcal{S}_* \subsetneq \mathcal{F}_*, \mathcal{S}_* \neq 0 \mid -C_{14} < \text{ch}_1(\mathcal{S}_*) \cdot [\eta^{n-1}]\}$. Then for any $\mathcal{S}_* \in \mathcal{A}$ and index i with $\lambda_i \neq 0$ we have $-C_{15} < \text{ch}_1(\mathcal{S}_*) \cdot [e_i] < C_{15}$. Recall that

$$\text{ch}_1(\mathcal{S}_*) = \text{ch}_1(\mathcal{S}) + \sum_{i \in I} \sum_{a \in [0, 1]} a \cdot \text{rank}_{D_i}({}^i \text{Gr}_a \mathcal{S}_*) \cdot [D_i].$$

Since $\mathcal{A}$ is a collection of parabolic subsheaves, as $\mathcal{S}_*$ varies in $\mathcal{A}$, the nontrivial parabolic weights a in the above expression take values from a finite set of real numbers. As $\text{ch}_1(\mathcal{S})$ and $[D_i]$ belong to $H^2(X, \mathbb{Z})$ and $e_i \in H^{2(n-1)}(X, \mathbb{Q})$, we see that $\text{ch}_1(\mathcal{S}_*) \cdot e_i$ can only achieve finite number of values as we vary $\mathcal{S}_* \in \mathcal{A}$ and the index i whence so does the values of $\text{ch}_1(\mathcal{S}_*) \cdot [\eta^{n-1}]$. The proof is completed. $\square$

Let $\mathcal{F}_*$ be a saturated reflexive parabolic sheaf which is semistable with respect to a big class $[\eta] \in \mathcal{B}^1$. It can be proved as in the ordinary sheaf case that there exists a Jordan-Hölder filtration

$$0 = \mathcal{F}_{0*} \subset \mathcal{F}_{1*} \subset \mathcal{F}_{2*} \subset \cdots \subset \mathcal{F}_{n*} = \mathcal{F}_*$$

where for each $0 \leq i < n$, $\mathcal{F}_{i*}$ is a saturated reflexive parabolic sheaf, $\text{Gr}_i \mathcal{F}_* := \mathcal{F}_{i+1*}/\mathcal{F}_{i*}$ is a η -stable parabolic sheaf with $\mu_\eta(\text{Gr}_i \mathcal{F}_*) = \mu_\eta(\mathcal{F}_*)$.

Then it follows directly from Lemma 7.2 that there exists an ϵ_0 such that for any $0 < \epsilon < \epsilon_0$ the quotient $\mathcal{F}_{i+1*}/\mathcal{F}_{i*}$ is η_ϵ -stable.

Proof of Theorem 7.1 $\mathcal{F}_{0*}$ is a saturated reflexive parabolic sheaf that is stable with respect to η_ϵ . Under the assumption that η is nef and big, we know that η_ϵ is a Kähler metric. Then we have $\Delta(\mathcal{F}_{0*}) \cdot [\eta_\epsilon^{n-1}] \geq 0$ whence $\Delta(\mathcal{F}_{0*}) \cdot [\eta^{n-1}] \geq 0$ by taking limit. On the other hand, Lemma 2.9 implies that the reflexive saturation $\text{Gr}_i \mathcal{F}'_*$ of $\text{Gr}_i \mathcal{F}_*$ is η_ϵ -stable. Then it follows from Lemma 2.8 that

$$\Delta(\text{Gr}_i \mathcal{F}_*) \cdot [\eta^{n-1}] \geq \Delta(\text{Gr}_i \mathcal{F}'_*) \cdot [\eta^{n-1}] \geq 0.$$

Hence it suffices to show that if we have the short exact sequence of parabolic sheaves

$$0 \longrightarrow \mathcal{F}_{i*} \longrightarrow \mathcal{F}_{i+1*} \longrightarrow \text{Gr}_i \mathcal{F}_* \longrightarrow 0,$$

with $\Delta(\mathcal{F}_{i*}) \cdot [\eta^{n-1}] \geq 0$ and $\Delta(\text{Gr}_i \mathcal{F}_*) \cdot [\eta^{n-1}] \geq 0$, we can obtain that $\Delta(\mathcal{F}_{i+1*}) \cdot [\eta^{n-1}] \geq 0$. But this is a standard fact, cf. [9, Lemma 3.7]. $\square$

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References

1. Bando, S.: Removable Singularities for Holomorphic Vector Bundles. *Tohoku Math. J.* **43**(1), 61–67 (1991). <https://doi.org/10.2748/tmj/1178227535>
2. Bando, S., Siu, Y.-T.: Stable Sheaves and Einstein-Hermitian Metrics. *Geom. Anal. Complex Manifolds* **39**, 39–50 (1994). https://doi.org/10.1142/9789814350112_0002
3. Biquard, O.: Fibrés de Higgs et Connexions Intégrables: Le Cas Logarithmique (Diviseur Lisse). French. In: *Ann. Sci. Éc. Norm. Sup. 30* (1997), pp. 41–96. https://www.numdam.org/article/ASENS_1997_4_30_1_41_0.pdf
4. Biquard, O.: Sur les Fibrés Paraboliques sur une Surface Complexe. *J. Lond. Math. Soc.* **53**(2), 302–316 (1996). <https://doi.org/10.1112/jlms/53.2.302>
5. Biswas, I.: Parabolic Bundles as Orbifold Bundles. *Duke Math. J.* **88**(2), 305–325 (1997). <https://doi.org/10.1215/S0012-7094-97-08812-8>
6. Borne, N.: Fibrés Paraboliques et Champ des Racines. In: *arXiv Math.* (2006). Preprint at <https://arxiv.org/abs/math/0604458>, [math/0604458](https://arxiv.org/abs/math/0604458)
7. Cao, J.-Y.: A Remark on Compact Kähler Manifolds with Nef Anticanonical Bundles and Its Applications. Preprint at [arxiv:1305.4397](https://arxiv.org/abs/1305.4397). (2013)

8. Cheng, S.-Y., Li, P.: Heat Kernel Estimates and Lower Bound of Eigenvalues. *Comment. Math. Helv.* **56**(1), 327–338 (1981). <https://doi.org/10.1007/BF01214381>
9. Claudon, B., Graf, P., Guenancia, H.: Numerical Characterization of Complex Torus Quotients. *Comment. Math. Helv.* **97**(4), 769–799 (2022). <https://doi.org/10.4171/CMH/543>
10. Donaldson, S.K.: A New Proof of a Theorem of Narasimhan and Seshadri. *J. Differ. Geom.* **18**, 279–315 (1983). <https://doi.org/10.4310/jdg/1214437664>
11. Donaldson, S.K.: Anti Self-Dual Yang-Mills Connections over Complex Algebraic Surfaces and Stable Vector Bundles. *Proc. Lond. Math. Soc.* **50**(3), 1–26 (1985). <https://doi.org/10.1112/plms/s3-50.1.1>
12. Grigor'yan, A.: Gaussian Upper Bounds for the Heat Kernel on Arbitrary Manifolds. *J. Differ. Geom.* **45**(1), 33–52 (1997). <https://doi.org/10.4310/jdg/1214459753>
13. Grivaux, J.: Chern classes in Deligne cohomology for coherent analytic sheaves. *Math. Ann.* **347**(2), 249–284 (2010)
14. Guenancia, H., Păun, M.: Conic Singularities Metrics with Prescribed Ricci Curvature: General Cone Angles along Normal Crossing Divisors. *J. Differ. Geom.* **103**(1), 15–57 (2016). <https://doi.org/10.4310/jdg/1460463562>
15. Guo, B.: et al. Sobolev Inequalities on Kähler Spaces. Preprint at [arxiv:2311.00221](https://arxiv.org/abs/2311.00221). (2023)
16. Hitchin, N.J.: The Self-Duality Equations on a Riemann Surface. *Proc. Lond. Math. Soc.* **55**, 59–126 (1987). <https://doi.org/10.1112/plms/s3-55.1.59>
17. Iyer, J.N., Simpson, C.T.: The Chern Character of a Parabolic Bundle, and a Parabolic Corollary of Reznikov's Theorem. In: *Geom. Dyn. Groups Spaces* (2008), pp. 439–485. https://doi.org/10.1007/978-3-7643-8608-5_12
18. Jost, J., Zuo, K.: Harmonic Maps of Infinite Energy and Rigidity Results for Representations of Fundamental Groups of Quasiprojective Varieties. *J. Differ. Geom.* **47**(3), 469–503 (1997). <https://doi.org/10.4310/jdg/1214459973>
19. Kaufmann, L., Lärkäng, R., Wulcan, E.: Global Chern Currents of Coherent Sheaves and Baum-Bott Currents. Preprint at [arxiv:2404.14585](https://arxiv.org/abs/2404.14585). 2024
20. Kobayashi, S.: Curvature and Stability of Vector Bundles. *Proc. Japan Acad. Ser. A* **58**, 158–162 (1982). <https://doi.org/10.3792/pjaa.58.158>
21. Kobayashi, S.: First Chern Class and Holomorphic Tensor Fields. *Nagoya Math. J.* **77**, 5–11 (1980). <https://doi.org/10.1017/S0027763000018602>
22. Li, J.-Y.: Hermitian-Einstein Metrics and Chern Number Inequalities on Parabolic Stable Bundles over Kähler Manifolds. *Commun. Anal. Geom.* **8**(3), 445–475 (2000)
23. Li, J.-Y., Narasimhan, M.S.: Hermitian-Einstein Metrics on Parabolic Stable Bundles. *Acta Math. Sin. Engl. Ser.* **15**(1), 93–114 (1999)
24. Li, J.-Y., Zhang, C.-J., Zhang, X.: Semi-Stable Higgs Sheaves and Bogomolov Type Inequality. *Calc. Var. Partial. Differ. Equ.* **56**(3), 81 (2017). <https://doi.org/10.1007/s00526-017-1187-4>
25. Lübke, M.: Stability of Einstein-Hermitian Vector Bundles. *Manuscripta Math.* **42**, 245–257 (1983). <https://doi.org/10.1007/BF01169586>
26. Maruyama, M., Yokogawa, K.: Moduli of Parabolic Stable Sheaves. *Math. Ann.* **293**, 77–99 (1992). <https://doi.org/10.1007/BF01444704>
27. Metha, V., Seshadri, C.S.: Moduli of Vector Bundles on Curves with Parabolic Structures. *Math. Ann.* **248**(3), 205–239 (1980). https://doi.org/10.1007/978-3-662-49105-2_3
28. Mochizuki, T.: Asymptotic Behaviour of Tame Harmonic Bundles and an Application to Pure Twistor D -Modules. *arXiv Math.* (2003). Preprint at <https://arxiv.org/abs/math/0312230>, math/0312230
29. Mochizuki, T.: Kobayashi-Hitchin Correspondence for Tame Harmonic Bundles and an Application. *French. In: Ast é risque* 309 (2006). https://www.numdam.org/item/AST_2006__309__R1_0/
30. Narasimhan, M.S., Seshadri, C.S.: Stable and Unitary Vector Bundles on Compact Riemann Surfaces. *Ann. Math.* **82**, 540–567 (1965). <https://doi.org/10.2307/1970710>
31. Sibner, L.M., Sibner, R.J.: Classification of Singular Sobolev Connections by Their Holonomy. *Commun. Math. Phys.* **144**(2), 337–350 (1992). <https://doi.org/10.1007/BF02101096>
32. Simpson, C.T.: Constructing Variation of Hodge Structure Using Yang-Mills Theory and Applications to Uniformization. *J. Amer. Math. Soc.* **1**(4), 867–918 (1988). <https://doi.org/10.1090/S0894-0347-1988-0944577-9>
33. Simpson, C.T.: Harmonic Bundles on Non-Compact Curves. *J. Amer. Math. Soc.* **3**, 713–770 (1990). <https://doi.org/10.1090/S0894-0347-1990-1040197-8>
34. Siu, Y.-T., Trautmann, G.: Gap-Sheaves and Extension of Coherent Analytic Subsheaves. Vol. 172. *Lect. Notes Math.* Berlin: Springer, 1971. <https://doi.org/10.1007/BFb0060577>
35. Steer, B., Wren, A.: The Donaldson-Hitchin-Kobayashi Correspondence for Parabolic Bundles over Orbifold Surfaces. *Can. J. Math.* **53**(6), 1309–1339 (2001). <https://doi.org/10.4153/CJM-2001-047-x>

36. Taher, C.H.: Calculating the Parabolic Chern Character of a Locally Abelian Parabolic Bundle. *Manuscr. Math.* **132**, 169–198 (2010). <https://doi.org/10.48550/arXiv.0904.0927>
37. Uhlenbeck, K.K., Yau, S.-T.: On the Existence of Hermitian-Yang-Mills Connections in Stable Vector Bundles. *Commun. Pure Appl. Math.* **39**, 257–293 (1986). <https://doi.org/10.1002/cpa.3160390714>

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